

BEACON EDUCATIONAL SERIES

Mathematics for Basic 8

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 8 (JHS 2) Students and Teachers

Beacon Educational Consult
2026 Edition

Mathematics for Basic 8
First Edition, 2026

Copyright © 2026 Beacon Educational Consult

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without prior written permission from the publisher.

This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

Published by: Beacon Educational Consult

ISBN: [To be assigned]

Printed in Ghana

PREFACE

This book has been developed to support Junior High School Basic 8 (JHS 2) students and teachers in their study of Mathematics, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

CONTENTS

Preface	iii
----------------	------------

CHAPTER MOCK EXAMINATIONS

Ch 1: Ch10	
Ch 2: Ch11	
Ch 3: Ch12	
Ch 4: Ch13	
Ch 5: Ch14	
Ch 6: Ch15	
Ch 7: Ch16	
Ch 8: Ch17	
Ch 9: Ch1	
Ch 10: Ch2	
Ch 11: Ch3	
Ch 12: Ch4	
Ch 13: Ch5	
Ch 14: Ch6	
Ch 15: Ch7	
Ch 16: Ch8	
Ch 17: Ch9	

ANSWER KEY

Chapter 1: Ch10

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Angles on a straight line add up to
 - A. 90°
 - B. 180°
 - C. 360°
 - D. 270°
2. The sum of angles at a point is
 - A. 180°
 - B. 90°
 - C. 270°
 - D. 360°
3. Two straight lines intersect and one of the angles formed is 70° . The vertically opposite angle is
 - A. 70°
 - B. 110°
 - C. 90°
 - D. 180°
4. Two angles on a straight line are x° and 115° . The value of x is
 - A. 55°
 - B. 75°
 - C. 65°
 - D. 115°
5. Angles around a point are 120° , 85° , 90° and x° . The value of x is
 - A. 75°
 - B. 65°
 - C. 55°
 - D. 95°
6. The supplement of an angle is three times the angle. The angle is
 - A. 60°
 - B. 30°
 - C. 45°
 - D. 90°

- 7.** The complement of 38° is
- A. 52°
 - B. 42°
 - C. 62°
 - D. 142°
- 8.** Two vertically opposite angles are $(2x + 10)^\circ$ and 70° . The value of x is
- A. 40
 - B. 25
 - C. 35
 - D. 30
- 9.** When a transversal cuts two parallel lines, the angles in matching positions are called
- A. alternate angles
 - B. corresponding angles
 - C. co-interior angles
 - D. vertically opposite angles
- 10.** A transversal cuts two parallel lines. If one of a pair of corresponding angles is 72° , the other is
- A. 72°
 - B. 108°
 - C. 18°
 - D. 90°
- 11.** A transversal cuts two parallel lines. If one of a pair of alternate angles is 55° , the other is
- A. 125°
 - B. 35°
 - C. 45°
 - D. 55°
- 12.** When a transversal cuts two parallel lines, a pair of co-interior angles is
- A. equal
 - B. supplementary
 - C. complementary
 - D. vertically opposite
- 13.** A transversal cuts two parallel lines. If one of a pair of co-interior angles is 124° , the other is
- A. 124°
 - B. 66°
 - C. 56°
 - D. 46°

- 14.** A transversal cuts two parallel lines. One of a pair of co-interior angles is three times the other. The smaller angle is
- A. 45°
 - B. 60°
 - C. 135°
 - D. 30°
- 15.** A transversal cuts two parallel lines. If one of the eight angles formed is 118° , the number of the eight angles equal to 118° is
- A. 2
 - B. 8
 - C. 6
 - D. 4
- 16.** A transversal cuts two parallel lines. Two angles on a straight line are y° and 132° . If y° is alternate to x° , then $x =$
- A. 132°
 - B. 58°
 - C. 48°
 - D. 42°
- 17.** A transversal cuts two parallel lines. If the difference between a pair of co-interior angles is 40° , the smaller angle is
- A. 70°
 - B. 110°
 - C. 80°
 - D. 40°
- 18.** A transversal cuts two parallel lines. One angle is $2x^\circ$ and its alternate angle is $(x + 50)^\circ$. The value of x is
- A. 25
 - B. 50
 - C. 100
 - D. 75
- 19.** A transversal cuts two parallel lines. If one angle is 37° , the supplement of its corresponding angle is
- A. 37°
 - B. 53°
 - C. 127°
 - D. 143°

- 20.** Which of the following shows that two lines cut by a transversal are parallel?
- A. Co-interior angles are equal
 - B. Alternate angles are supplementary
 - C. Corresponding angles are equal
 - D. Vertically opposite angles are equal
- 21.** The sum of the angles in a triangle is
- A. 360°
 - B. 180°
 - C. 90°
 - D. 270°
- 22.** In triangle ABC, $\angle A = 50^\circ$ and $\angle B = 60^\circ$. $\angle C =$
- A. 80°
 - B. 60°
 - C. 110°
 - D. 70°
- 23.** In triangle PQR, $\angle P = 2x^\circ$, $\angle Q = 3x^\circ$ and $\angle R = 4x^\circ$. The value of x is
- A. 20
 - B. 18
 - C. 25
 - D. 30
- 24.** The exterior angle of a triangle is equal to
- A. 180°
 - B. the third interior angle
 - C. the sum of the two interior opposite angles
 - D. 360°
- 25.** Two interior angles of a triangle are 65° and 48° . The exterior angle at the third vertex is
- A. 67°
 - B. 113°
 - C. 123°
 - D. 97°
- 26.** The sum of the interior angles of a polygon of n sides is
- A. $(n - 2) \times 180^\circ$
 - B. $n \times 180^\circ$
 - C. $(n + 2) \times 180^\circ$
 - D. 360°

- 27.** The sum of the interior angles of a hexagon is
- A. 540°
 - B. 900°
 - C. 720°
 - D. 630°
- 28.** Each interior angle of a regular hexagon is
- A. 108°
 - B. 135°
 - C. 60°
 - D. 120°
- 29.** The sum of the interior angles of a pentagon is
- A. 720°
 - B. 540°
 - C. 450°
 - D. 360°
- 30.** The sum of the exterior angles of any convex polygon is
- A. 360°
 - B. 180°
 - C. $(n - 2) \times 180^\circ$
 - D. 270°
- 31.** Each exterior angle of a regular octagon is
- A. 135°
 - B. 60°
 - C. 36°
 - D. 45°
- 32.** A regular polygon has an interior angle of 135° . The number of sides of the polygon is
- A. 6
 - B. 10
 - C. 8
 - D. 9
- 33.** Each interior angle of a regular polygon is 144° . The polygon is a
- A. decagon
 - B. nonagon
 - C. octagon
 - D. dodecagon

- 34.** The angles of a triangle are x° , $(x + 20)^\circ$ and $(x + 40)^\circ$. The smallest angle is
- A. 60°
 - B. 40°
 - C. 80°
 - D. 50°
- 35.** In triangle ABC, $AB = AC$ and $\angle B = 68^\circ$. $\angle A =$
- A. 68°
 - B. 112°
 - C. 22°
 - D. 44°
- 36.** In an isosceles triangle, the vertical angle is 40° . Each base angle is
- A. 140°
 - B. 70°
 - C. 40°
 - D. 100°
- 37.** The angles of a triangle are in the ratio 2 : 3 : 4. The greatest angle is
- A. 60°
 - B. 90°
 - C. 80°
 - D. 40°
- 38.** A transversal cuts two parallel lines. One angle is three times its adjacent angle on the same straight line. The smaller angle is
- A. 45°
 - B. 60°
 - C. 135°
 - D. 30°
- 39.** The sum of the interior angles of a polygon is 1440° . The number of sides of the polygon is
- A. 8
 - B. 12
 - C. 9
 - D. 10
- 40.** In a regular polygon, the ratio of an interior angle to its exterior angle is 4 : 1. The number of sides of the polygon is
- A. 8
 - B. 10
 - C. 12
 - D. 9

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Angle facts.

Question 2

(a) Transversals.

Question 3

(a) Angles in triangles.

Question 4

(a) Angles in polygons.

Question 5

(a) Applications.

Question 6

(a) Reasoning and deduction.

Chapter 2: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following angles cannot be constructed using a pair of compasses and a ruler only?
 - A. 75°
 - B. 80°
 - C. 105°
 - D. 135°
2. The first step in constructing an angle of 60° with a pair of compasses is to
 - A. draw a ray and an arc with the vertex as centre
 - B. draw a perpendicular to the ray
 - C. bisect a right angle
 - D. measure the angle with a protractor
3. To bisect an angle means to
 - A. divide it into three equal angles
 - B. double the angle
 - C. divide it into two equal angles
 - D. make it a right angle
4. When an angle of 135° is bisected, each of the two angles formed is
 - A. 45°
 - B. 70°
 - C. 65°
 - D. $67\frac{1}{2}^\circ$
5. Bisecting an angle of 150° gives two angles each of
 - A. 75°
 - B. 50°
 - C. 80°
 - D. 60°
6. In constructing an angle of 120° with a pair of compasses, after drawing an arc with the vertex as centre, the next step is to
 - A. draw a line through the vertex
 - B. measure the angle with a protractor
 - C. draw a perpendicular to the ray

- D. use the same radius to step off two equal arcs along the first arc
- 7.** To bisect an angle, arcs are drawn from the two points where the first arc cuts the arms of the angle. These two arcs must have
- A. different radii
 - B. radii equal to the lengths of the arms
 - C. equal radii
 - D. radii equal to the size of the angle
- 8.** An angle of 135° can be constructed as the sum of
- A. 100° and 35°
 - B. 90° and 45°
 - C. 80° and 55°
 - D. 120° and 25°
- 9.** The instrument used to check the size of a constructed angle of 105° is
- A. a ruler
 - B. a pair of set squares only
 - C. a divider
 - D. a protractor
- 10.** An angle of 75° can be constructed by bisecting the angle between rays making
- A. 60° and 90°
 - B. 30° and 90°
 - C. 45° and 90°
 - D. 60° and 120°
- 11.** To construct an equilateral triangle, it is enough to know
- A. one angle
 - B. two sides
 - C. two angles
 - D. one side
- 12.** An equilateral triangle is constructed with side 6.2 cm. Each angle of the triangle is
- A. 45°
 - B. 90°
 - C. 60°
 - D. 120°
- 13.** To construct a triangle when all three sides are given, the construction is done by
- A. using a protractor at each end of the base
 - B. drawing arcs from both ends of the base with radii equal to the other two sides

- C. bisecting the base
 - D. drawing perpendiculars at the ends of the base
- 14.** A triangle with sides 6 cm, 8 cm and 7 cm is
- A. scalene
 - B. isosceles
 - C. equilateral
 - D. right-angled
- 15.** To construct an isosceles triangle with base 6 cm and base angles 50° each, the step after drawing the base is to
- A. bisect the base
 - B. draw an arc of radius 6 cm
 - C. construct 50° at each end of the base
 - D. construct 80° at one end of the base
- 16.** The vertical angle of the isosceles triangle in question 15 is
- A. 50°
 - B. 100°
 - C. 40°
 - D. 80°
- 17.** A triangle with angles 35° , 55° and 90° is classified as
- A. acute-angled
 - B. right-angled
 - C. obtuse-angled
 - D. equilateral
- 18.** Which of the following sets of angles determines an obtuse-angled triangle?
- A. 25° , 35° , 120°
 - B. 40° , 60° , 80°
 - C. 30° , 60° , 90°
 - D. 50° , 50° , 80°
- 19.** In constructing a triangle given two sides and the included angle, the angle is constructed
- A. opposite the longer side
 - B. after drawing all three sides
 - C. at any convenient position
 - D. between the two given sides

- 20.** In triangle ABC, $\angle A = 45^\circ$, $AC = 3$ cm and $AB = 5$ cm. After constructing the angle and marking the two sides, the third side BC is obtained by
- A. measuring it with a ruler only
 - B. joining B and C with a straight line
 - C. bisecting angle A
 - D. drawing a circle with centre A
- 21.** Which of the following is enough to construct a unique triangle?
- A. Three angles only
 - B. One side and one angle
 - C. Three sides
 - D. Two angles only
- 22.** Three angles only (AAA) determine
- A. infinitely many triangles of different sizes
 - B. a unique triangle
 - C. no triangle at all
 - D. a right-angled triangle only
- 23.** For three line segments to form a triangle, the sum of the lengths of any two of them must be
- A. equal to the third
 - B. less than the third
 - C. twice the third
 - D. greater than the third
- 24.** Which of the following sets of lengths cannot form a triangle?
- A. 6 cm, 7 cm, 8 cm
 - B. 3 cm, 4 cm, 8 cm
 - C. 3 cm, 4 cm, 5 cm
 - D. 5 cm, 5 cm, 6 cm
- 25.** The locus of a point which is always 5 cm from a fixed point O is
- A. a straight line
 - B. a circle of radius 10 cm
 - C. a circle with centre O and radius 5 cm
 - D. a ray from O
- 26.** The locus of a point equidistant from two fixed points P and Q is
- A. the perpendicular bisector of PQ
 - B. the line PQ
 - C. a circle with diameter PQ
 - D. the angle bisector at P

- 27.** The locus of points equidistant from two intersecting straight lines is
- A. a pair of parallel lines
 - B. a circle
 - C. the perpendicular bisector of the lines
 - D. the pair of angle bisectors of the angles between the lines
- 28.** The locus of points equidistant from two parallel lines is
- A. a line perpendicular to both lines
 - B. a circle between them
 - C. a line midway between and parallel to both lines
 - D. two intersecting lines
- 29.** The centre of a drawn circle can be located by constructing
- A. the angle bisectors of two chords
 - B. the perpendicular bisectors of two chords
 - C. lines through any two points on the circle
 - D. the perpendicular bisectors of two radii
- 30.** A goat is tied to a post with a rope 4 m long. The region the goat can graze is
- A. a circle of radius 4 m with the post as centre
 - B. a circle of diameter 4 m
 - C. a square of side 4 m
 - D. a straight line 4 m long
- 31.** Two points P and Q are 5 cm apart. The number of points which are 3 cm from both P and Q is
- A. 0
 - B. 2
 - C. 1
 - D. 4
- 32.** The point equidistant from the three vertices of a triangle is found at the intersection of the
- A. angle bisectors
 - B. medians
 - C. altitudes
 - D. perpendicular bisectors of the sides
- 33.** The point equidistant from the three sides of a triangle is found at the intersection of the
- A. angle bisectors
 - B. perpendicular bisectors
 - C. medians
 - D. diagonals

- 34.** The locus of points at a fixed distance from a given straight line is
- A. one parallel line
 - B. a circle
 - C. two parallel lines, one on each side of the given line
 - D. a perpendicular line
- 35.** A fountain is equidistant from two straight paths which meet at a point. The fountain lies on
- A. the perpendicular bisector of the paths
 - B. the angle bisector of the angle between the paths
 - C. a circle with centre at the meeting point
 - D. the longer path
- 36.** A tree is 10 m from a flagpole and also 10 m from a lamp post which is 12 m from the flagpole. Its position is found by
- A. drawing arcs of radius 10 m from both points
 - B. bisecting the line joining the two points
 - C. drawing parallel lines through the two points
 - D. drawing one circle of radius 20 m
- 37.** Kofi walks so that he is always 6 m from a long straight wall. His path is
- A. perpendicular to the wall
 - B. a circle of radius 6 m
 - C. parallel to the wall and 6 m from it
 - D. along the wall
- 38.** An angle of 150° can be constructed as the sum of
- A. 120° and 45°
 - B. 60° and 45°
 - C. 100° and 50°
 - D. 90° and 60°
- 39.** A triangle has sides 5 cm, 5 cm and 6 cm. It is best described as
- A. scalene
 - B. isosceles
 - C. equilateral
 - D. right-angled
- 40.** A construction combines the locus of points 5 cm from a fixed point O with the locus of points equidistant from two fixed points A and B. The required points are
- A. where the circle with centre O meets the perpendicular bisector of AB
 - B. a single circle
 - C. the angle bisectors of A and B
 - D. two parallel lines

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Construction of angles.

Question 2

(a) More angle constructions.

Question 3

(a) Construction of triangles.

Question 4

(a) More triangle constructions.

Question 5

(a) Loci.

Question 6

(a) Integrated loci.

Chapter 3: Ch12

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The ratio of the circumference of any circle to its diameter is
 - A. π
 - B. 2π
 - C. r
 - D. $2r$
2. The circumference of a circle of radius 7 cm (take $\pi = 22/7$) is
 - A. 154 cm
 - B. 44 cm
 - C. 22 cm
 - D. 308 cm
3. The area of a circle of radius 14 cm (take $\pi = 22/7$) is
 - A. 88 cm^2
 - B. 154 cm^2
 - C. 616 cm^2
 - D. 1232 cm^2
4. The area of a semi-circle of radius 7 cm (take $\pi = 22/7$) is
 - A. 77 cm^2
 - B. 154 cm^2
 - C. 38.5 cm^2
 - D. 44 cm^2
5. Two circles have a common centre; the smaller has radius 7 cm and the bigger has radius 14 cm. The area of the region between them (take $\pi = 22/7$) is
 - A. 154 cm^2
 - B. 308 cm^2
 - C. 462 cm^2
 - D. 616 cm^2
6. The formula for the area of a circle is deduced by cutting the circle into many sectors and arranging them to form a
 - A. rectangle
 - B. triangle
 - C. trapezium

D. parallelogram

- 7.** In the sector method of deducing the area of a circle, the rectangle formed has length and width approximately
- A. $2\pi r$ and r
 - B. πr and r
 - C. πr and $2r$
 - D. r^2 and r
- 8.** The circumference of a circle of diameter 10 cm (take $\pi = 3.14$) is
- A. 62.8 cm
 - B. 15.7 cm
 - C. 78.5 cm
 - D. 31.4 cm
- 9.** The radius of a circle whose circumference is 88 cm (take $\pi = 22/7$) is
- A. 28 cm
 - B. 7 cm
 - C. 14 cm
 - D. 44 cm
- 10.** The area of a circle whose circumference is 44 cm (take $\pi = 22/7$) is
- A. 44 cm^2
 - B. 154 cm^2
 - C. 616 cm^2
 - D. 88 cm^2
- 11.** A circular garden of radius 7 m is to be fenced at GH¢2.00 per metre (take $\pi = 22/7$). The cost of the fencing is
- A. GH¢44.00
 - B. GH¢154.00
 - C. GH¢88.00
 - D. GH¢308.00
- 12.** The area of a circle is 154 cm^2 (take $\pi = 22/7$). Its radius is
- A. 7 cm
 - B. 14 cm
 - C. 49 cm
 - D. 22 cm
- 13.** In a right-angled triangle with hypotenuse c and the other two sides a and b ,
- A. $a + b = c$
 - B. $a^2 + b^2 = c^2$
 - C. $a^2 - b^2 = c^2$
 - D. $2a + 2b = 2c$

- 14.** The two shorter sides of a right-angled triangle are 3 cm and 4 cm. The hypotenuse is
- A. 7 cm
 - B. 12 cm
 - C. 5 cm
 - D. 25 cm
- 15.** A right-angled triangle has hypotenuse 13 cm and one side 5 cm. The third side is
- A. 18 cm
 - B. 8 cm
 - C. 144 cm
 - D. 12 cm
- 16.** A ladder leans against a vertical wall of height 12 m. The foot of the ladder is 5 m from the wall. The length of the ladder is
- A. 17 m
 - B. 13 m
 - C. 7 m
 - D. $\sqrt{119}$ m
- 17.** A ladder leans against a vertical wall of height 8 m. The foot of the ladder is 6 m from the wall. The length of the ladder is
- A. 10 m
 - B. 14 m
 - C. 48 m
 - D. 100 m
- 18.** A boat travels 2 m South and then 9 m East. Its distance from the starting point, correct to one decimal place, is
- A. 11.0 m
 - B. 8.1 m
 - C. 7.0 m
 - D. 9.2 m
- 19.** An isosceles triangle has equal sides 10 cm long and a base 12 cm long. The altitude of the triangle is
- A. 6 cm
 - B. 8 cm
 - C. $\sqrt{136}$ cm
 - D. 16 cm

- 20.** The side of an equilateral triangle is 12 cm. The height of the triangle, correct to one decimal place, is
- A. 6.0 cm
 - B. 12.0 cm
 - C. 36.0 cm
 - D. 10.4 cm
- 21.** The area of the equilateral triangle in question 20, correct to one decimal place, is
- A. 72.0 cm²
 - B. 62.4 cm²
 - C. 124.8 cm²
 - D. 36.0 cm²
- 22.** The perimeter of the equilateral triangle in question 20 is
- A. 24 cm
 - B. 48 cm
 - C. 36 cm
 - D. 144 cm
- 23.** Yeboah hangs a picture frame of width 15 cm on the wall with a string passing over a nail. The distance from the nail to each edge of the frame is 10 cm. The length of the string is
- A. 25 cm
 - B. 10 cm
 - C. 15 cm
 - D. 20 cm
- 24.** The vertical distance from the nail to the top edge of the frame in question 23, correct to one decimal place, is
- A. 6.6 cm
 - B. 7.5 cm
 - C. 10.0 cm
 - D. 2.5 cm
- 25.** Which of the following is a Pythagorean triple?
- A. 5, 12, 14
 - B. 6, 7, 10
 - C. 8, 15, 17
 - D. 9, 12, 16

- 26.** A right-angled triangle has sides 6 cm and 8 cm. The area of the square on its hypotenuse is
- A. 48 cm^2
 - B. 100 cm^2
 - C. 14 cm^2
 - D. 96 cm^2
- 27.** The diagonal of a rectangle of length 12 cm and width 5 cm is
- A. 17 cm
 - B. 60 cm
 - C. 7 cm
 - D. 13 cm
- 28.** The hypotenuse of a right-angled triangle is 25 cm and one side is 7 cm. The third side is
- A. 24 cm
 - B. 18 cm
 - C. 32 cm
 - D. 576 cm
- 29.** In a right-angled triangle, $\sin \theta =$
- A. adjacent $\div$ hypotenuse
 - B. opposite $\div$ adjacent
 - C. opposite $\div$ hypotenuse
 - D. hypotenuse $\div$ opposite
- 30.** In a right-angled triangle, $\tan \theta =$
- A. adjacent $\div$ hypotenuse
 - B. opposite $\div$ adjacent
 - C. hypotenuse $\div$ opposite
 - D. opposite $\div$ hypotenuse
- 31.** A right-angled triangle has sides 3, 4 and 5, with the hypotenuse 5. If θ is the angle opposite the side 3, then $\sin \theta =$
- A. $4/5$
 - B. $3/4$
 - C. $3/5$
 - D. $5/3$
- 32.** For the triangle in question 31, $\cos \theta =$
- A. $3/5$
 - B. $5/4$
 - C. $3/4$
 - D. $4/5$

- 33.** For the triangle in question 31, $\tan \theta =$
- A. $3/4$
 - B. $4/3$
 - C. $3/5$
 - D. $4/5$
- 34.** The angle which the line of sight makes with the horizontal when an observer looks up at an object is the
- A. angle of depression
 - B. angle of elevation
 - C. right angle
 - D. angle of contact
- 35.** A hunter on top of a tower of height 18 m sees a fire at an angle of depression of 30° . The distance from the hunter to the fire is
- A. 9 m
 - B. 18 m
 - C. 36 m
 - D. 54 m
- 36.** A right-angled triangle has hypotenuse 20 cm. If $\sin \theta = 3/5$, the side opposite θ is
- A. 12 cm
 - B. 16 cm
 - C. 15 cm
 - D. 9 cm
- 37.** A circle of radius 7 cm rolls once along a straight line. The distance it travels (take $\pi = 22/7$) is
- A. 154 cm
 - B. 44 cm
 - C. 22 cm
 - D. 7 cm
- 38.** A ladder 10 m long reaches 8 m up a vertical wall. The distance of its foot from the wall is
- A. 2 m
 - B. 18 m
 - C. 4 m
 - D. 6 m

- 39.** The area of a circle is 616 cm^2 (take $\pi = 22/7$). Its circumference is
- A. 44 cm
 - B. 154 cm
 - C. 88 cm
 - D. 28 cm
- 40.** Kofi walks 8 m North from a flagpole and then 15 m East. His distance from the flagpole is
- A. 17 m
 - B. 23 m
 - C. 7 m
 - D. 120 m

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) The circle.

Question 2

(a) The Pythagoras theorem.

Question 3

(a) Pythagoras in real life.

Question 4

(a) Areas of triangles using Pythagoras.

Question 5

(a) Trigonometric ratios.

Question 6

(a) Integrated problems.

Chapter 4: Ch13

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A quantity which has both magnitude and direction is called
 - A. a scalar
 - B. a magnitude
 - C. a vector
 - D. a component
2. The vector from point A to point B is written as
 - A. vector AB
 - B. vector BA
 - C. scalar AB
 - D. magnitude AB
3. A vector with zero magnitude is called the
 - A. unit vector
 - B. equal vector
 - C. parallel vector
 - D. zero vector
4. If $p = (2, -1)$, then $-p =$
 - A. $(2, 1)$
 - B. $(-2, 1)$
 - C. $(-2, -1)$
 - D. $(1, 2)$
5. The vector $(3, 4)$ represents a movement of
 - A. 3 units to the right and 4 units up
 - B. 4 units to the right and 3 units up
 - C. 3 units to the left and 4 units up
 - D. 3 units up and 4 units to the right
6. The vector $(-2, 5)$ represents a movement of
 - A. 2 units to the right and 5 units up
 - B. 2 units to the left and 5 units down
 - C. 5 units to the left and 2 units up
 - D. 2 units to the left and 5 units up

- 7.** If $a = (2, 3)$ and $b = (4, 1)$, then $a + b =$
- A. $(8, 3)$
 - B. $(6, 3)$
 - C. $(6, 4)$
 - D. $(2, 12)$
- 8.** If $a = (5, -2)$ and $b = (-3, 4)$, then $a + b =$
- A. $(2, -6)$
 - B. $(2, 2)$
 - C. $(8, 2)$
 - D. $(-15, -8)$
- 9.** If $a = (-1, -4)$ and $b = (-2, 3)$, then $a + b =$
- A. $(-3, -1)$
 - B. $(1, -7)$
 - C. $(-3, 1)$
 - D. $(3, 1)$
- 10.** If $AB = (3, 2)$ and $BC = (1, 5)$, then $AC =$
- A. $(2, -3)$
 - B. $(3, 10)$
 - C. $(7, 4)$
 - D. $(4, 7)$
- 11.** If $PQ = (-2, 6)$ and $QR = (5, -1)$, then $PR =$
- A. $(7, -7)$
 - B. $(3, 5)$
 - C. $(-3, -5)$
 - D. $(10, -6)$
- 12.** If $a + (2, -3) = (5, 4)$, then $a =$
- A. $(7, 1)$
 - B. $(3, 1)$
 - C. $(3, 7)$
 - D. $(-3, -7)$
- 13.** If $XY = (4, -1)$ and $XZ = (6, 3)$, then $YZ =$
- A. $(2, 4)$
 - B. $(10, 2)$
 - C. $(-2, -4)$
 - D. $(2, -4)$

- 14.** If $a = (1, 2)$, $b = (3, -1)$ and $c = (-2, 4)$, then $a + b + c =$
- A. $(6, 5)$
 - B. $(2, -3)$
 - C. $(4, 3)$
 - D. $(2, 5)$
- 15.** If $a = (4, -3)$, then $2a =$
- A. $(6, -1)$
 - B. $(8, -6)$
 - C. $(8, 6)$
 - D. $(2, -5)$
- 16.** If $p = (2, -1)$, then $3p =$
- A. $(6, -3)$
 - B. $(5, 2)$
 - C. $(6, 3)$
 - D. $(-6, -3)$
- 17.** If $q = (3, 4)$, then $-2q =$
- A. $(1, 2)$
 - B. $(6, 8)$
 - C. $(-6, -8)$
 - D. $(-6, 8)$
- 18.** If $p = (2, -1)$ and $q = (3, 4)$, then $3q - 2p =$
- A. $(13, 10)$
 - B. $(5, 14)$
 - C. $(5, 10)$
 - D. $(-1, -14)$
- 19.** If $p = (2, -1)$ and $r = (-2, 3)$, then $r - 3p =$
- A. $(4, 0)$
 - B. $(-8, -6)$
 - C. $(4, 6)$
 - D. $(-8, 6)$
- 20.** If $p = (2, -1)$, $q = (3, 4)$ and $r = (-2, 3)$, then $q - p + 2r =$
- A. $(-3, 11)$
 - B. $(1, 9)$
 - C. $(-3, 5)$
 - D. $(7, 1)$

- 21.** If $a = (-3, 2)$, then $-a + 2a =$
- A. $(-6, 4)$
 - B. $(0, 0)$
 - C. $(-9, 6)$
 - D. $(-3, 2)$
- 22.** If $a = (4, -6)$, then $\frac{1}{2}a =$
- A. $(4, -3)$
 - B. $(2, -3)$
 - C. $(8, -12)$
 - D. $(2, 3)$
- 23.** If $k(x, y) = (12, -8)$ and $k = 4$, then $(x, y) =$
- A. $(3, -2)$
 - B. $(16, -4)$
 - C. $(48, -32)$
 - D. $(-3, 2)$
- 24.** If $a = (2, 1)$ and $b = (1, -3)$, then $2a - 3b =$
- A. $(1, -7)$
 - B. $(7, 11)$
 - C. $(1, 11)$
 - D. $(-1, -11)$
- 25.** If $AB = (4, 2)$, then $BA =$
- A. $(4, 2)$
 - B. $(2, 4)$
 - C. $(4, -2)$
 - D. $(-4, -2)$
- 26.** If $a = (-2, 5)$ and $b = (3, -1)$, then $b - a =$
- A. $(1, 4)$
 - B. $(-5, 6)$
 - C. $(5, -6)$
 - D. $(-1, -4)$
- 27.** Two vectors are equal if and only if they have the same
- A. starting point
 - B. magnitude and direction
 - C. magnitude only
 - D. direction only

28. If $(x, 4) = (7, y)$, then

- A. $x = 7, y = 4$
- B. $x = 4, y = 7$
- C. $x = 7, y = -4$
- D. $x = -7, y = 4$

29. If $(2x, 3) = (8, y)$, then $x + y =$

- A. 11
- B. 5
- C. 6
- D. 7

30. If $(x + 2, y - 1) = (5, 3)$, then

- A. $x = 7, y = 2$
- B. $x = 3, y = 2$
- C. $x = 3, y = 4$
- D. $x = 2, y = 3$

31. If $2(x, y) = (6, -10)$, then $(x, y) =$

- A. $(3, -5)$
- B. $(8, -8)$
- C. $(12, -20)$
- D. $(-3, 5)$

32. If $a = (3, -2)$ and $b = (x + 1, y)$, and $a = b$, then

- A. $x = 3, y = -2$
- B. $x = 4, y = -2$
- C. $x = 2, y = 2$
- D. $x = 2, y = -2$

33. A man walks 3 km East and then 4 km North. The magnitude of his resultant displacement is

- A. 7 km
- B. 5 km
- C. 12 km
- D. 25 km

34. A boat sails $(2, 5)$ km and then $(-1, 3)$ km. Its resultant displacement is

- A. $(3, 2)$
- B. $(1, 2)$
- C. $(1, 8)$
- D. $(-2, 15)$

- 35.** A girl's total displacement after two walks is $(6, -2)$. If the first walk was $(2, 3)$, the second walk was
- A. $(4, -5)$
 - B. $(8, 1)$
 - C. $(-4, 5)$
 - D. $(4, 5)$
- 36.** If $AB = (a, b)$ and $BC = (c, d)$, then $AC =$
- A. $(a - c, b - d)$
 - B. $(a + c, b + d)$
 - C. (ac, bd)
 - D. $(a + c, b - d)$
- 37.** If $AB = (a, b)$ and $BC = (c, d)$, then $AB - BC =$
- A. $(a + c, b + d)$
 - B. $(c - a, d - b)$
 - C. $(a - c, d - b)$
 - D. $(a - c, b - d)$
- 38.** The position of B relative to A is given by the vector $(5, -3)$. The position of A relative to B is
- A. $(5, 3)$
 - B. $(3, -5)$
 - C. $(-5, 3)$
 - D. $(-5, -3)$
- 39.** If $k(3, -2) = (-9, 6)$, the value of k is
- A. -3
 - B. 3
 - C. -6
 - D. $-1/3$
- 40.** If $a = (2, 4)$ and $b = (1, 2)$, then
- A. $a = b$
 - B. $a = 2b$
 - C. $b = 2a$
 - D. $a + b = 0$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Vector concepts.

Question 2

(a) Addition of vectors.

Question 3

(a) Subtraction and scalar multiplication.

Question 4

(a) Combined operations.

Question 5

(a) Vector equality.

Question 6

(a) Applications.

Chapter 5: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following is an example of rotation in everyday life?
 - A. a book sliding along a table
 - B. a stone falling from a height
 - C. a child walking along a straight path
 - D. the blades of a turning fan
2. A rotation is fully described by its
 - A. centre and distance only
 - B. centre, angle and direction
 - C. angle and length only
 - D. direction and distance only
3. The fixed point about which a shape turns is called the
 - A. centre of rotation
 - B. angle of rotation
 - C. line of symmetry
 - D. point of origin
4. The hands of a clock move
 - A. anticlockwise
 - B. in both directions
 - C. clockwise
 - D. along a straight line
5. A quarter turn corresponds to an angle of
 - A. 45°
 - B. 180°
 - C. 360°
 - D. 90°
6. A half turn corresponds to an angle of
 - A. 90°
 - B. 180°
 - C. 270°
 - D. 360°

- 7.** A complete turn corresponds to an angle of
- A. 90°
 - B. 180°
 - C. 360°
 - D. 270°
- 8.** A steering wheel of a car and the blades of a ceiling fan both demonstrate
- A. rotation
 - B. translation
 - C. reflection
 - D. enlargement
- 9.** The image of the point $(2, 3)$ under a rotation of 90° anticlockwise about the origin is
- A. $(3, 2)$
 - B. $(-2, -3)$
 - C. $(3, -2)$
 - D. $(-3, 2)$
- 10.** The image of the point $(4, -1)$ under a rotation of 90° anticlockwise about the origin is
- A. $(-1, -4)$
 - B. $(1, 4)$
 - C. $(-4, -1)$
 - D. $(1, -4)$
- 11.** The image of the point $(5, 2)$ under a rotation of 90° clockwise about the origin is
- A. $(2, -5)$
 - B. $(-2, 5)$
 - C. $(-5, -2)$
 - D. $(5, -2)$
- 12.** The image of the point $(-3, 4)$ under a rotation of 180° about the origin is
- A. $(-3, -4)$
 - B. $(4, 3)$
 - C. $(-4, -3)$
 - D. $(3, -4)$
- 13.** The image of the point $(-2, -5)$ under a rotation of 180° about the origin is
- A. $(-5, -2)$
 - B. $(5, 2)$
 - C. $(2, 5)$
 - D. $(-2, 5)$

- 14.** The image of the point $(6, -3)$ under a rotation of 270° anticlockwise about the origin is
- A. $(3, 6)$
 - B. $(-3, -6)$
 - C. $(-6, 3)$
 - D. $(6, 3)$
- 15.** The image of the point $(-1, -4)$ under a rotation of 90° anticlockwise about the origin is
- A. $(4, -1)$
 - B. $(-4, 1)$
 - C. $(1, 4)$
 - D. $(-4, -1)$
- 16.** A point P is rotated through 360° about the origin. Its image is
- A. the reflection of P
 - B. the point opposite P through the origin
 - C. P itself
 - D. the origin
- 17.** Under a rotation of 90° anticlockwise about the origin, the rule is $(x, y) \rightarrow$
- A. $(y, -x)$
 - B. $(-y, x)$
 - C. $(-x, -y)$
 - D. (y, x)
- 18.** Under a rotation of 180° about the origin, the rule is $(x, y) \rightarrow$
- A. (y, x)
 - B. $(-y, x)$
 - C. $(y, -x)$
 - D. $(-x, -y)$
- 19.** The point $(3, 1)$ is rotated 90° clockwise about the origin. Its image lies in the
- A. fourth quadrant
 - B. first quadrant
 - C. second quadrant
 - D. third quadrant
- 20.** The point $(-2, 5)$ is rotated 180° about the origin. Its image lies in the
- A. first quadrant
 - B. second quadrant
 - C. fourth quadrant
 - D. third quadrant

- 21.** The vertices of triangle ABC are A(1, 0), B(3, 0) and C(1, 2). Under a rotation of 90° anticlockwise about the origin, the image of C is
- A. (2, 1)
 - B. (-1, -2)
 - C. (2, -1)
 - D. (-2, 1)
- 22.** The image of the point (a, b) under a rotation of 90° clockwise about the origin is
- A. (b, -a)
 - B. (-b, a)
 - C. (-a, -b)
 - D. (b, a)
- 23.** Under a rotation about the origin, the point (2, 7) maps onto (-7, 2). The rotation is
- A. 90° clockwise
 - B. 90° anticlockwise
 - C. 180°
 - D. 270° anticlockwise
- 24.** Under a rotation about the origin, the point (3, -2) maps onto (-3, 2). The rotation is
- A. 90° clockwise
 - B. 90° anticlockwise
 - C. 180°
 - D. 360°
- 25.** The angle of rotation is found by joining the centre of rotation to a point and its image, then measuring the angle between
- A. the point and the origin
 - B. the image and the x-axis
 - C. the object and the y-axis
 - D. the two joining lines
- 26.** A wheel turns through three quarter-turns. The total angle of rotation is
- A. 180°
 - B. 270°
 - C. 90°
 - D. 360°
- 27.** The minute hand of a clock moves from 12 to 3. The angle and direction of rotation are
- A. 90° clockwise
 - B. 90° anticlockwise
 - C. 180° clockwise

D. 270° clockwise

- 28.** The minute hand of a clock moves from 12 to 8. The clockwise angle through which it turns is
- A. 120°
 - B. 160°
 - C. 200°
 - D. 240°
- 29.** A shape is rotated so that a point and its image subtend a right angle at the centre of rotation. The angle of rotation is
- A. 45°
 - B. 180°
 - C. 90°
 - D. 360°
- 30.** A fan turns through 540° . The number of complete turns made is
- A. $1\frac{1}{4}$
 - B. $1\frac{1}{2}$
 - C. 2
 - D. $\frac{3}{4}$
- 31.** The point (2, 0) is rotated about the origin onto (0, 2). The anticlockwise angle of rotation is
- A. 90°
 - B. 180°
 - C. 270°
 - D. 45°
- 32.** The point (3, 0) is rotated about the origin onto (0, -3). The anticlockwise angle of rotation is
- A. 90°
 - B. 180°
 - C. 270°
 - D. 360°
- 33.** After a rotation, the image of a shape is always
- A. larger than the object
 - B. congruent to the object
 - C. smaller than the object
 - D. a different shape

- 34.** Under a rotation, corresponding sides of the object and its image are
- A. parallel
 - B. perpendicular
 - C. doubled
 - D. equal in length
- 35.** Which of the following is preserved under a rotation?
- A. the lengths of sides and the sizes of angles
 - B. the coordinates of the vertices
 - C. the position of the shape
 - D. the direction the shape faces
- 36.** Two shapes are congruent if they
- A. have the same area only
 - B. are enlargements of each other
 - C. have exactly the same shape and size
 - D. have the same perimeter only
- 37.** Triangle PQR is rotated onto triangle P'Q'R'. If $PQ = 5$ cm, then $P'Q' =$
- A. 2.5 cm
 - B. 10 cm
 - C. 25 cm
 - D. 5 cm
- 38.** In question 37, if $\angle P = 40^\circ$, then $\angle P' =$
- A. 40°
 - B. 50°
 - C. 80°
 - D. 140°
- 39.** Which of the following transformations always maps a shape onto a congruent image?
- A. enlargement
 - B. rotation
 - C. stretching
 - D. shrinking
- 40.** Under a rotation, which of the following changes?
- A. the lengths of the sides
 - B. the sizes of the angles
 - C. the position of the shape
 - D. the area of the shape

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Rotation concepts.

Question 2

(a) Rotation rules about the origin.

Question 3

(a) Rotating a shape.

Question 4

(a) Angle of rotation.

Question 5

(a) Congruence.

Question 6

(a) Integrated problems.

Chapter 6: Ch15

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Data which can be counted and takes whole-number values only is called
 - A. discrete data
 - B. continuous data
 - C. categorical data
 - D. grouped data
2. Which of the following is an example of continuous data?
 - A. the number of children in a family
 - B. the number of learners in a class
 - C. the type of movie watched
 - D. the weights of babies in a crèche
3. Which of the following is an example of discrete data?
 - A. the height of a student
 - B. the number of cars sold in a year
 - C. the time taken to run a race
 - D. the amount of rainfall in a month
4. Which of the following is categorical (qualitative) data?
 - A. the number of learners in a B8 class
 - B. scores in a class test
 - C. the marital status of a person
 - D. the heights of students
5. Data collected directly, before it is organised or processed, is called
 - A. raw data
 - B. grouped data
 - C. secondary data
 - D. information
6. Data arranged into classes or intervals is called
 - A. raw data
 - B. discrete data
 - C. grouped data
 - D. categorical data

- 7.** Data which can take any value within a given range is called
- A. discrete data
 - B. continuous data
 - C. categorical data
 - D. qualitative data
- 8.** 'Sex' (male or female) is an example of
- A. categorical data
 - B. continuous data
 - C. discrete data
 - D. grouped data
- 9.** The scores of 11 learners in a class test are 25, 30, 35, 40, 45, 26, 29, 50, 45, 37 and 47. These scores are best described as
- A. categorical data
 - B. grouped data
 - C. continuous data
 - D. ungrouped numerical (discrete) data
- 10.** Boxers' weight classes and age groups are examples of non-numeric data which can be placed on
- A. a pie chart only
 - B. a number line
 - C. an ordinal scale
 - D. a tally chart only
- 11.** The best method of collecting data on the favourite colours of learners in a school is
- A. an experiment
 - B. a questionnaire
 - C. weighing
 - D. measuring
- 12.** To find out whether eating twice a person's normal number of cream crackers increases productivity, the best method of data collection is
- A. an experiment
 - B. observation only
 - C. a pie chart
 - D. a pictogram
- 13.** Ama wants to know how people in her community obtain drinking water. The most appropriate method of data collection is
- A. measuring their heights
 - B. weighing them
 - C. using a stem-and-leaf plot

- D. an interview (questionnaire)
- 14.** Data collected by watching and recording events without changing anything is collected by
- A. experiment
 - B. questionnaire
 - C. observation
 - D. simulation
- 15.** A printed form designed to collect information from respondents is called
- A. a bar chart
 - B. a questionnaire
 - C. a tally chart
 - D. a waffle chart
- 16.** Which of the following questions is suitable for a quantitative study?
- A. What is your favourite song?
 - B. Which movie do you enjoy most?
 - C. What is your religion?
 - D. How many hours do you study each day?
- 17.** A tally chart is used to
- A. measure heights
 - B. calculate averages
 - C. record frequencies using strokes
 - D. draw circles
- 18.** The tally group of four vertical strokes crossed by a diagonal stroke represents
- A. 4
 - B. 5
 - C. 9
 - D. 10
- 19.** The lengths, in millimetres, of 40 leaves were measured to the nearest mm and grouped into classes 25-29, 30-34, 35-39, 40-44, 45-49, 50-54 and 55-59. If the class 25-29 contains the leaves 25 mm and 28 mm only, its frequency is
- A. 2
 - B. 5
 - C. 25
 - D. 28

- 20.** Using the leaf-length data, the frequency of the class 40–44 is
- A. 7
 - B. 8
 - C. 4
 - D. 9
- 21.** Using the leaf-length data, the modal class is
- A. 40–44
 - B. 25–29
 - C. 45–49
 - D. 55–59
- 22.** Using the leaf-length data, the number of leaves with lengths of at least 50 mm is
- A. 7
 - B. 10
 - C. 3
 - D. 13
- 23.** Using the leaf-length data, the total number of leaves measured is
- A. 36
 - B. 38
 - C. 39
 - D. 40
- 24.** The class width of the classes 25–29, 30–34 and 35–39 is
- A. 4
 - B. 29
 - C. 5
 - D. 25
- 25.** In a stem-and-leaf plot of test scores, the entry '4 | 7' represents the score
- A. 47
 - B. 74
 - C. 4.7
 - D. 407
- 26.** In a stem-and-leaf plot, if no score falls in the 30s, the row for stem 3 should be
- A. written with a 0 against it
 - B. filled with 3s
 - C. left blank
 - D. removed from the plot

- 27.** In a bar chart, the height of each bar represents the
- A. frequency
 - B. class width
 - C. tally
 - D. percentage only
- 28.** A bar chart is most suitable for displaying data which are
- A. only continuous
 - B. categorical or discrete
 - C. only grouped
 - D. only percentages
- 29.** The total angle at the centre of a complete pie chart is
- A. 90°
 - B. 180°
 - C. 270°
 - D. 360°
- 30.** A cleaner of a small office spent GH¢120.00 on food, GH¢80.00 on rent, GH¢40.00 on clothing, GH¢110.00 on transport and saved GH¢50.00. The total amount spent and saved is
- A. GH¢350.00
 - B. GH¢360.00
 - C. GH¢400.00
 - D. GH¢440.00
- 31.** From the cleaner's data in question 30, the sector angle for food in a pie chart is
- A. 72°
 - B. 108°
 - C. 120°
 - D. 99°
- 32.** From the cleaner's data in question 30, the sector angle for transport in a pie chart is
- A. 45°
 - B. 36°
 - C. 110°
 - D. 99°
- 33.** From the cleaner's data in question 30, the sector angle for clothing in a pie chart is
- A. 40°
 - B. 50°
 - C. 36°
 - D. 72°

- 34.** From the cleaner's data in question 30, the fraction of the total amount which was spent on rent is
- A. $\frac{1}{5}$
 - B. $\frac{1}{4}$
 - C. $\frac{2}{5}$
 - D. $\frac{1}{3}$
- 35.** In a pictogram, one symbol represents 5 learners. The number of symbols needed to represent 35 learners is
- A. 5
 - B. 7
 - C. 35
 - D. 175
- 36.** In a waffle chart of 100 cells, each cell represents
- A. 10%
 - B. 5%
 - C. 100%
 - D. 1%
- 37.** A waffle chart shows that the average score of B7 learners in a mathematics test is 64%. The number of shaded cells is
- A. 64
 - B. 36
 - C. 6.4
 - D. 100
- 38.** In a pictogram, one full symbol represents 10 mangoes. The number of mangoes represented by $3\frac{1}{2}$ symbols is
- A. 3.5
 - B. 30
 - C. 35
 - D. 70
- 39.** Which of the following displays is best for showing how a whole quantity is shared among its parts?
- A. a stem-and-leaf plot
 - B. a pie chart
 - C. a tally chart
 - D. a scatter diagram

- 40.** The data 12, 15, 15, 23, 27 and 31 are displayed in a stem-and-leaf plot. The number of leaves against stem 1 is
- A. 2
 - B. 6
 - C. 1
 - D. 3

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Types of data.

Question 2

(a) Methods of data collection.

Question 3

(a) Frequency tables.

Question 4

(a) Bar charts and pie charts.

Question 5

(a) Stem-and-leaf plots, waffle charts and pictograms.

Question 6

(a) Integrated problems.

Chapter 7: Ch16

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The mean of 4, 7, 9 and 12 is
 - A. 32
 - B. 8
 - C. 7
 - D. 10
2. The mean of 5, 5, 8 and 10 is
 - A. 28
 - B. 6
 - C. 7
 - D. 8
3. The sum of six numbers is 48. Their mean is
 - A. 8
 - B. 288
 - C. 6
 - D. 42
4. The mean of three numbers is 10. Two of the numbers are 8 and 13. The third number is
 - A. 10
 - B. 31
 - C. 11
 - D. 9
5. The mean of x , 5, 7 and 10 is 7. The value of x is
 - A. 5
 - B. 6
 - C. 7
 - D. 28
6. Ama's scores in five tests are 12, 15, 14, 16 and 13. Her mean score is
 - A. 14
 - B. 70
 - C. 13
 - D. 15

- 7.** The mean of 2.5, 3.5 and 6 is
- A. 3
 - B. 12
 - C. 3.5
 - D. 4
- 8.** Kojo's taxi made 8, 6, 10, 10, 9, 10 and 3 trips from Monday to Sunday. The mean number of trips per day is
- A. 56
 - B. 10
 - C. 8
 - D. 9
- 9.** The mean of a set of ten values is 6. If one value of 6 is removed, the mean of the remaining values is
- A. 5.4
 - B. 6
 - C. 6.6
 - D. cannot be determined
- 10.** The mean height of four pupils is 140 cm. When a fifth pupil joins the group, the mean becomes 142 cm. The height of the fifth pupil is
- A. 150 cm
 - B. 142 cm
 - C. 148 cm
 - D. 154 cm
- 11.** The mode of 2, 3, 3, 5, 7, 3 and 8 is
- A. 5
 - B. 8
 - C. 2
 - D. 3
- 12.** The median of 4, 9, 1, 7 and 5 is
- A. 4
 - B. 5
 - C. 7
 - D. 9
- 13.** The median of 2, 4, 6, 8, 10 and 12 is
- A. 7
 - B. 6
 - C. 8
 - D. 7.5

- 14.** The mode of 1, 2, 2, 3, 4, 4 and 5 is
- A. 2 only
 - B. 4 only
 - C. 2 and 4
 - D. there is no mode
- 15.** For the data set 3, 5, 5, 7, 9, which of the following is true?
- A. the mean equals the median
 - B. the median is less than the mode
 - C. the mean equals the mode
 - D. the median equals the mode
- 16.** The areas of the sitting-room floors of seven houses are 22, 24, 26, 30, 48, 30 and 30 m². The mean area is
- A. 26 m²
 - B. 210 m²
 - C. 30 m²
 - D. 48 m²
- 17.** For the house-area data in question 16, the median area is
- A. 30 m²
 - B. 26 m²
 - C. 48 m²
 - D. 24 m²
- 18.** For the house-area data in question 16, the mode is
- A. 22 m²
 - B. 26 m²
 - C. 48 m²
 - D. 30 m²
- 19.** For the house-area data in question 16, the mean, median and mode are all equal because
- A. the data contain an outlier
 - B. the data are balanced around 30
 - C. all seven values are equal
 - D. 48 appears only once
- 20.** In a frequency table, the data value 2 has the greatest frequency. The mode of the data is
- A. the greatest frequency
 - B. the sum of the frequencies
 - C. 2
 - D. 11

- 21.** The range of 4, 9, 2, 11 and 6 is
- A. 9
 - B. 32
 - C. 11
 - D. 2
- 22.** The range of Kojo's taxi trips in question 8 is
- A. 10
 - B. 7
 - C. 3
 - D. 56
- 23.** The greatest value in a data set is 25 and the range is 18. The smallest value is
- A. 43
 - B. 18
 - C. 7
 - D. 25
- 24.** The range of 5, 5, 5 and 5 is
- A. 5
 - B. 20
 - C. 1
 - D. 0
- 25.** The measure which uses only the greatest and the smallest values of a data set is the
- A. range
 - B. mean
 - C. median
 - D. mode
- 26.** The range of the house-area data in question 16 is
- A. 30 m^2
 - B. 22 m^2
 - C. 48 m^2
 - D. 26 m^2
- 27.** The measure of central tendency most affected by an extreme value (outlier) is the
- A. mode
 - B. median
 - C. mean
 - D. range

- 28.** The most appropriate measure of central tendency for finding the most popular shoe size in a shop is the
- A. mean
 - B. mode
 - C. median
 - D. range
- 29.** The most appropriate measure for describing a typical income in a community where a few people earn far more than the rest is the
- A. median
 - B. mean
 - C. mode
 - D. range
- 30.** The most appropriate measure to use when every value in the data set matters equally is the
- A. mode
 - B. range
 - C. median
 - D. mean
- 31.** The median of Kojo's taxi trips in question 8 is
- A. 8
 - B. 9
 - C. 10
 - D. 3
- 32.** The mode of Kojo's taxi trips in question 8 is
- A. 10
 - B. 9
 - C. 8
 - D. 56
- 33.** The measure which best represents the typical number of trips Kojo makes each day is
- A. the mean, because it uses all the values
 - B. the range
 - C. the mode, because 10 is the greatest value
 - D. the median, because it is not pulled down by the low value of 3
- 34.** In a class test most pupils scored around 15 marks, but one pupil scored 0 because she was absent. The teacher reporting the class performance should use
- A. the mean, because it is exact
 - B. the range
 - C. the median or the mode, because the 0 is an extreme value

D. any of the three, because they are all equally suitable

- 35.** A frequency table shows the values 1 to 7 with frequencies 1, 11, 5, 4, 2, 1 and 1. The total number of data values is
- A. 7
 - B. 25
 - C. 11
 - D. 77
- 36.** For the frequency table in question 35, the mean of the data is
- A. 3
 - B. 4
 - C. 3.08
 - D. 77
- 37.** For the frequency table in question 35, the median of the data is
- A. 2
 - B. 11
 - C. 4
 - D. 3
- 38.** For the frequency table in question 35, the mode of the data is
- A. 2
 - B. 11
 - C. 3
 - D. 1
- 39.** For the frequency table in question 35, the mean, median and mode are different because
- A. the data are symmetric
 - B. the data are skewed, with most values small
 - C. all the frequencies are equal
 - D. no value is repeated
- 40.** The mean, median and mode of a data set are all equal. The data set is most likely
- A. skewed to the left
 - B. skewed to the right
 - C. symmetric
 - D. full of outliers

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) The mean.

Question 2

- (a) Median, mode and range.

Question 3

- (a) The house-area data.

Question 4

- (a) Kojo's taxi trips.

Question 5

- (a) A frequency table.

Question 6

- (a) Choosing and creating.

Chapter 8: Ch1

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The set of all possible outcomes of a probability experiment is called the
 - A. event
 - B. probability
 - C. trial
 - D. sample space
2. The sample space for tossing a fair coin once is
 - A. {H, T}
 - B. {H}
 - C. {H, H, T}
 - D. {1, 2}
3. The sample space for throwing a fair six-sided die has how many outcomes?
 - A. 2
 - B. 3
 - C. 6
 - D. 12
4. Two coins are tossed once. The sample space is
 - A. {HH, TT}
 - B. {HH, HT, TH, TT}
 - C. {HH, HT, TT}
 - D. {H, T}
5. A bag contains 3 red and 2 green bottle tops. One bottle top is picked at random. The number of possible colour outcomes is
 - A. 1
 - B. 3
 - C. 5
 - D. 2
6. A die is thrown and a coin is tossed. The number of outcomes in the sample space is
 - A. 8
 - B. 12
 - C. 6

D. 36

- 7.** Two events are independent if the outcome of one
- A. does not affect the outcome of the other
 - B. determines the outcome of the other
 - C. is always equal to the outcome of the other
 - D. always happens before the other
- 8.** Ampofo walks to school daily and Serwaa travels to school by bus daily. The event 'Ampofo walks' and the event 'Serwaa travels by bus' are
- A. dependent events
 - B. the same event
 - C. impossible events
 - D. independent events
- 9.** A bottle top is picked from a bag, replaced, and a second bottle top is picked. The two picks are
- A. dependent
 - B. impossible
 - C. independent
 - D. certain
- 10.** A coin is tossed and a spinner numbered 1 and 2 is spun. The number of outcomes in the sample space is
- A. 2
 - B. 4
 - C. 6
 - D. 8
- 11.** A box contains 3 blue pens and 4 pink pens. A pen is taken, its colour noted and replaced, and another pen is taken. The sample space of the two trials is
- A. {BB, BP, PB, PP}
 - B. {BB, PP}
 - C. {B, P}
 - D. {BB, BP, PP}
- 12.** A spinner with 3 equal sections is spun twice. The number of possible outcomes is
- A. 3
 - B. 6
 - C. 9
 - D. 12

- 13.** A fair die is thrown. The probability of scoring a 6 is
- A. $\frac{1}{2}$
 - B. $\frac{1}{6}$
 - C. 6
 - D. $\frac{1}{3}$
- 14.** A fair die is thrown. The probability of scoring an even number is
- A. $\frac{1}{6}$
 - B. $\frac{1}{3}$
 - C. $\frac{2}{3}$
 - D. $\frac{1}{2}$
- 15.** A bag contains 3 red, 2 green and 1 pink bottle top. One bottle top is picked at random. The probability of picking a red bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{6}$
 - D. $\frac{2}{3}$
- 16.** From the bag in question 15, the probability of picking a green bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{6}$
 - C. $\frac{1}{3}$
 - D. $\frac{2}{5}$
- 17.** From the bag in question 15, the probability of picking the pink bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{4}$
 - D. $\frac{1}{6}$
- 18.** From the bag in question 15, the probability of NOT picking a red bottle top is
- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{6}$
 - D. $\frac{5}{6}$
- 19.** A box contains 3 blue pens and 4 pink pens. One pen is taken at random. The probability that it is blue is
- A. $\frac{3}{4}$
 - B. $\frac{3}{7}$
 - C. $\frac{4}{7}$
 - D. $\frac{1}{7}$

- 20.** From the box in question 19, the probability that the pen taken is pink is
- A. $\frac{3}{7}$
 - B. $\frac{4}{3}$
 - C. $\frac{4}{7}$
 - D. $\frac{1}{4}$
- 21.** A pen is taken from the box in question 19, replaced, and a second pen is taken. The probability that both pens are blue is
- A. $\frac{3}{7}$
 - B. $\frac{6}{7}$
 - C. $\frac{6}{49}$
 - D. $\frac{9}{49}$
- 22.** From the experiment in question 21, the probability that both pens are pink is
- A. $\frac{12}{49}$
 - B. $\frac{16}{49}$
 - C. $\frac{4}{7}$
 - D. $\frac{8}{49}$
- 23.** From the experiment in question 21, the probability of taking one blue pen and one pink pen, in any order, is
- A. $\frac{24}{49}$
 - B. $\frac{12}{49}$
 - C. $\frac{9}{49}$
 - D. $\frac{25}{49}$
- 24.** The probability of an event which is certain to happen is
- A. 0
 - B. $\frac{1}{2}$
 - C. 1
 - D. greater than 1
- 25.** The probability of an impossible event is
- A. 1
 - B. $\frac{1}{2}$
 - C. between 0 and 1
 - D. 0
- 26.** A spinner records a win or a loss each time it is spun. If $P(\text{win}) = \frac{3}{5}$, then $P(\text{loss}) =$
- A. $\frac{2}{5}$
 - B. $\frac{3}{5}$
 - C. $\frac{5}{3}$
 - D. $\frac{1}{5}$

- 27.** The probability $\frac{1}{4}$ expressed as a decimal is
- A. 0.4
 - B. 0.25
 - C. 4.0
 - D. 2.5
- 28.** The probability $\frac{1}{4}$ expressed as a percentage is
- A. 4%
 - B. 14%
 - C. 40%
 - D. 25%
- 29.** The probability $\frac{1}{4}$ expressed as a ratio of favourable outcomes to total outcomes is
- A. 1 : 3
 - B. 4 : 1
 - C. 1 : 4
 - D. 3 : 4
- 30.** The probability 0.6 expressed as a fraction in its lowest terms is
- A. $\frac{3}{5}$
 - B. $\frac{6}{10}$
 - C. $\frac{3}{10}$
 - D. $\frac{5}{3}$
- 31.** The probability 45% expressed as a fraction in its lowest terms is
- A. $\frac{45}{10}$
 - B. $\frac{9}{20}$
 - C. $\frac{4}{5}$
 - D. $\frac{9}{10}$
- 32.** The probability $\frac{3}{8}$ expressed as a decimal is
- A. 0.83
 - B. 3.8
 - C. 0.375
 - D. 0.308
- 33.** The probability of picking a red bottle top from the bag in question 15, expressed as a percentage, is
- A. 50%
 - B. 30%
 - C. 3%
 - D. 500%

- 34.** An event has probability $\frac{2}{7}$. The ratio of favourable outcomes to total outcomes is
- A. 7 : 2
 - B. 2 : 5
 - C. 5 : 7
 - D. 2 : 7
- 35.** A spinner with two equal sections, Win and Lose, is spun twice. The probability of winning twice is
- A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
 - C. $\frac{3}{4}$
 - D. $\frac{1}{8}$
- 36.** From the spinner in question 35, the probability of losing twice is
- A. $\frac{1}{2}$
 - B. $\frac{3}{4}$
 - C. $\frac{1}{4}$
 - D. 0
- 37.** From the spinner in question 35, the probability of exactly one win in the two spins is
- A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
 - C. $\frac{3}{4}$
 - D. 1
- 38.** In a probability tree diagram, the probability of two successive events is found by
- A. adding the probabilities on the branches
 - B. multiplying the probabilities on the branches
 - C. subtracting the probabilities on the branches
 - D. dividing the probabilities on the branches
- 39.** In a complete probability tree diagram, the sum of the probabilities of all the final outcomes is
- A. 0
 - B. $\frac{1}{2}$
 - C. 2
 - D. 1
- 40.** A die is thrown at most three times, and the game stops when a 6 is scored. The probability that the game stops at the first throw is
- A. $\frac{1}{3}$
 - B. $\frac{1}{2}$
 - C. $\frac{1}{6}$

D. 6

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Sample spaces.

Question 2

- (a) The bottle-top experiment.

Question 3

- (a) The pen experiment.

Question 4

- (a) The spinner experiment.

Question 5

- (a) Conversions.

Question 6

- (a) Integrated problems.

Chapter 9: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The place value of the digit 6 in the number 2,468,321,507 is
 - A. 6,000,000
 - B. 600,000
 - C. 60,000,000
 - D. 6,000
2. The number 3,000,000,000 written in words is
 - A. three billion
 - B. three million
 - C. thirty billion
 - D. three hundred million
3. Two million, four hundred and eight thousand, three hundred and twenty-one written in figures is
 - A. 2,480,321
 - B. 2,408,231
 - C. 2,048,321
 - D. 2,408,321
4. The value of the digit 5 in the numeral part of the serial number TD1,567,451 is
 - A. 5,000
 - B. 500,000
 - C. 50,000
 - D. 5,000,000
5. How many millions make one billion?
 - A. 1,000
 - B. 100
 - C. 10
 - D. 1,000,000

- 6.** The figure 4,050,006,000 written in words is
- A. four billion, five million and six thousand
 - B. four billion, fifty million and six thousand
 - C. forty billion, five million and sixty thousand
 - D. four billion, fifty million and six
- 7.** The number of zeros in one billion is
- A. 6
 - B. 8
 - C. 9
 - D. 12
- 8.** In the number 7,891,234,560, the digit in the hundred millions place is
- A. 8
 - B. 9
 - C. 7
 - D. 1
- 9.** Counting forwards in 500,000s beginning from 200,000, the 3rd number is
- A. 700,000
 - B. 1,000,000
 - C. 1,500,000
 - D. 1,200,000
- 10.** Counting backwards in 100,050s beginning from 1,800,000, the 3rd number is
- A. 1,700,000
 - B. 1,699,950
 - C. 1,599,900
 - D. 1,500,000
- 11.** Counting forwards in 500,000s, the number that comes immediately after 2,500,000 is
- A. 2,550,000
 - B. 3,000,000
 - C. 3,500,000
 - D. 2,000,000
- 12.** Counting backwards in 10,000s beginning from 1,000,000, the 5th number is
- A. 960,000
 - B. 950,000
 - C. 970,000
 - D. 940,000

- 13.** In a skip count, the number after 750,000 is 800,000. The counting is done in
- A. 5,000s
 - B. 100,000s
 - C. 50,000s
 - D. 25,000s
- 14.** Which symbol makes the statement true? $3,450,000,000 \square 3,405,000,000$
- A. $<$
 - B. $=$
 - C. $\leq$
 - D. $>$
- 15.** Which of the following numbers is the greatest?
- A. 9,999,999
 - B. 1,000,000,001
 - C. 1,000,000,000
 - D. 999,999,999
- 16.** Which of the following numbers is the smallest?
- A. 2,009,500,000
 - B. 2,090,500,000
 - C. 2,900,050,000
 - D. 2,095,000,000
- 17.** The numbers 4,050,000,000; 4,005,000,000 and 4,500,000,000 arranged in ascending order are
- A. 4,050,000,000; 4,005,000,000; 4,500,000,000
 - B. 4,500,000,000; 4,050,000,000; 4,005,000,000
 - C. 4,005,000,000; 4,500,000,000; 4,050,000,000
 - D. 4,005,000,000; 4,050,000,000; 4,500,000,000
- 18.** The number that is 500,000 more than 1,295,300,000 is
- A. 1,295,350,000
 - B. 1,300,300,000
 - C. 1,295,800,000
 - D. 1,296,300,000
- 19.** Which of the following numbers lies between 2,400,000,000 and 2,500,000,000?
- A. 2,390,000,000
 - B. 2,450,000,000
 - C. 2,510,000,000
 - D. 2,550,000,000

20. The standard form of 1,000 is

- A. 10×10^2
- B. 1×10^2
- C. 0.1×10^4
- D. 1×10^3

21. The standard form of 26 is

- A. 2.6×10
- B. 2.6×10^2
- C. 0.26×10^2
- D. 26×10

22. The standard form of 375 is

- A. 37.5×10
- B. 3.75×10^3
- C. 3.75×10^2
- D. 0.375×10^3

23. The standard form of 8,650,049 is

- A. 8.650049×10^5
- B. 8.650049×10^6
- C. 86.50049×10^5
- D. 8.650049×10^7

24. The standard form of 100 is

- A. 1×10^2
- B. 1×10^3
- C. 10×10
- D. 0.1×10^3

25. Written as an ordinary number, 4.2×10^5 is

- A. 42,000
- B. 4,200,000
- C. 4,200
- D. 420,000

26. The standard form of 1 is

- A. 1×10
- B. 0.1×10
- C. 1×10^0
- D. 10×10

27. Which of the following is NOT in standard form?

- A. 3.5×10^4
- B. 35×10^3
- C. 1.2×10^6
- D. 9.99×10^2

28. The standard form of 10 is

- A. 1×10
- B. 10×10^2
- C. 0.1×10^2
- D. 1×10^2

29. 56,734 expressed correct to two significant figures is

- A. 56,000
- B. 56,700
- C. 57,000
- D. 60,000

30. 975.8674 expressed correct to two decimal places is

- A. 975.86
- B. 975.87
- C. 975.9
- D. 976

31. 975.8674 expressed correct to three decimal places is

- A. 975.868
- B. 975.86
- C. 975.87
- D. 975.867

32. 56,734 expressed correct to three significant figures is

- A. 56,700
- B. 57,000
- C. 56,000
- D. 56,730

33. The number of significant figures in 0.04506 is

- A. 3
- B. 5
- C. 4
- D. 6

- 34.** 48,296 rounded to the nearest thousand is
- A. 49,000
 - B. 48,000
 - C. 48,300
 - D. 50,000
- 35.** 3,456,789 expressed correct to one significant figure is
- A. 3,400,000
 - B. 3,500,000
 - C. 4,000,000
 - D. 3,000,000
- 36.** Adom earns GH¢2,500 a month after tax and his elder brother Arko earns three times as much. If there is no increase in their earnings, their total income after five years is
- A. GH¢600,000
 - B. GH¢150,000
 - C. GH¢300,000
 - D. GH¢60,000
- 37.** The population of Ghana in a certain year was 30,832,019. Written in words, this is
- A. three million, eight hundred and thirty-two thousand and nineteen
 - B. thirty million, eight hundred and twenty-three thousand and nineteen
 - C. thirty million, eight hundred and thirty-two thousand and nineteen
 - D. thirty billion, eight hundred and thirty-two thousand and nineteen
- 38.** A company makes a profit of GH¢1,250,000 in a year. Written in standard form, this profit is
- A. 12.5×10^5
 - B. 1.25×10^6
 - C. 1.25×10^5
 - D. 0.125×10^7
- 39.** The digit in the ten-thousands place of 3,764,528 is replaced by 0. The new number is
- A. 3,764,028
 - B. 3,760,528
 - C. 3,064,528
 - D. 3,704,528
- 40.** The distance from Accra to Tamale is about 615 km. A bus makes the journey 1,000 times in a year. The total distance travelled in a year, in standard form, is
- A. 6.15×10^5 km
 - B. 6.15×10^6 km

C. $61.5 \times 10^4 \text{ km}$

D. $6.15 \times 10^3 \text{ km}$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Reading, writing and place value.

Question 2

(a) Skip counting.

Question 3

(a) Comparing and ordering.

Question 4

(a) Standard form.

Question 5

(a) Significant figures and decimal places.

Question 6

(a) Real-life problems.

Chapter 10: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The set of factors of 36 is
 - A. {1, 2, 3, 6, 12, 18, 36}
 - B. {2, 3, 4, 6, 9, 12, 18, 36}
 - C. {1, 2, 3, 4, 6, 9, 12, 18, 36}
 - D. {1, 2, 3, 4, 6, 8, 9, 12, 18, 36}
2. The number of elements in the set of factors of 42 is
 - A. 8
 - B. 6
 - C. 7
 - D. 9
3. The common factors of 42 and 36 are
 - A. {1, 2, 3}
 - B. {1, 3, 6, 9}
 - C. {2, 3, 6, 7}
 - D. {1, 2, 3, 6}
4. The highest common factor (HCF) of 42 and 36 is
 - A. 3
 - B. 6
 - C. 12
 - D. 2
5. Which of the following numbers is a factor of every whole number?
 - A. 1
 - B. 2
 - C. 3
 - D. 10
6. The set of common factors of 24 and 32 is
 - A. {1, 2, 4}
 - B. {1, 2, 4, 8}
 - C. {2, 4, 8, 16}
 - D. {1, 2, 3, 4, 6, 8}

- 7.** The largest factor of 120 which is also a factor of 45 is
- A. 5
 - B. 9
 - C. 15
 - D. 45
- 8.** Which of the following numbers is a perfect square?
- A. 144
 - B. 140
 - C. 48
 - D. 72
- 9.** Which of the following numbers is NOT a perfect square?
- A. 81
 - B. 100
 - C. 121
 - D. 110
- 10.** The perfect squares among the numbers 16, 24, 32, 36, 48 and 64 are
- A. 16, 24 and 36
 - B. 16, 32 and 64
 - C. 16, 36 and 64
 - D. 24, 36 and 48
- 11.** The smallest perfect square greater than 100 is
- A. 110
 - B. 121
 - C. 144
 - D. 101
- 12.** A perfect square is a number which
- A. has a whole number as its square root
 - B. is always even
 - C. has exactly two factors
 - D. is divisible by 4
- 13.** Which of the following sets contains only perfect squares?
- A. {4, 9, 18, 25}
 - B. {1, 4, 8, 16}
 - C. {1, 4, 9, 16}
 - D. {9, 16, 27, 36}

14. The next perfect square after 49 is

- A. 56
- B. 54
- C. 50
- D. 64

15. $\sqrt{81} =$

- A. 8
- B. 9
- C. 18
- D. 4.5

16. $\sqrt{169} =$

- A. 13
- B. 17
- C. 12
- D. 15

17. To find $\sqrt{49}$, consecutive odd numbers starting from 1 are subtracted from 49 until the remainder is zero. The number of subtractions gives $\sqrt{49}$, which is

- A. 5
- B. 6
- C. 49
- D. 7

18. $\sqrt{400} =$

- A. 40
- B. 200
- C. 20
- D. 4

19. $\sqrt{(36 \times 49)} =$

- A. 85
- B. 42
- C. 1,764
- D. 12

20. The value of $\sqrt{225} + \sqrt{64}$ is

- A. 17
- B. 289
- C. 15
- D. 23

- 21.** A square plot of land has an area of $10,000 \text{ m}^2$. The length of each side of the plot is
- A. 100 m
 - B. 1,000 m
 - C. 5,000 m
 - D. 50 m
- 22.** Which of the following numbers has a whole number as its square root?
- A. 90
 - B. 150
 - C. 196
 - D. 200
- 23.** The set of elements common to two sets is called their
- A. union
 - B. intersection
 - C. complement
 - D. universal set
- 24.** The union of two sets A and B is the set of
- A. all elements in A or B or both
 - B. elements common to A and B
 - C. elements in A but not in B
 - D. elements in neither A nor B
- 25.** If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then $A \cap B$ is
- A. $\{1, 2, 5, 6\}$
 - B. $\{1, 2, 3, 4, 5, 6\}$
 - C. $\{1, 2\}$
 - D. $\{3, 4\}$
- 26.** If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then $A \cup B$ is
- A. $\{3, 4\}$
 - B. $\{1, 2, 5, 6\}$
 - C. $\{1, 2, 3, 4, 5, 6\}$
 - D. $\{1, 2, 3, 4\}$
- 27.** If two sets have no element in common, then their intersection is
- A. the union of the two sets
 - B. the empty set $\{\}$
 - C. one of the sets
 - D. the universal set

- 28.** The symbol $\cap$ represents
- A. intersection
 - B. union
 - C. subset
 - D. complement
- 29.** If $A = \{2, 4, 6, 8, 10\}$ and $B = \{3, 6, 9, 12\}$, then $A \cap B$ is
- A. $\{3, 6\}$
 - B. $\{2, 4, 8, 10\}$
 - C. $\{6\}$
 - D. $\{\}$
- 30.** The number of elements in $A \cup B$, if $A = \{2, 4, 6, 8, 10\}$ and $B = \{3, 6, 9, 12\}$, is
- A. 9
 - B. 8
 - C. 7
 - D. 6
- 31.** In a class, M is the set of students who like Mathematics and E the set of students who like English. The students who like both subjects are represented by
- A. $M \cup E$
 - B. $E - M$
 - C. $M - E$
 - D. $M \cap E$
- 32.** If $n(A) = 12$, $n(B) = 15$ and $n(A \cap B) = 5$, then $n(A \cup B)$ is
- A. 22
 - B. 27
 - C. 17
 - D. 32
- 33.** In a Venn diagram of two sets A and B, the region where the two circles overlap represents
- A. $A \cup B$
 - B. the elements in A only
 - C. $A \cap B$
 - D. the elements in neither set
- 34.** In a village, 50 farmers grow maize and 60 grow rice. If all the 80 farmers grow at least one of the two crops, the number of farmers who grow both crops is
- A. 10
 - B. 30

- C. 50
D. 20
- 35.** In a class of 40 students, 25 play football, 20 play volleyball and every student plays at least one of the games. The number of students who play both games is
- A. 10
B. 15
C. 20
D. 5
- 36.** In a class of 40 students, 25 play football, 20 play volleyball and 5 play both games. The number of students who play football only is
- A. 20
B. 25
C. 15
D. 30
- 37.** In a survey of 60 people, 35 own a radio, 30 own a television and 10 own neither. The number of people who own both a radio and a television is
- A. 10
B. 25
C. 15
D. 20
- 38.** If $P = \{\text{factors of } 12\}$ and $Q = \{\text{factors of } 18\}$, then $P \cap Q$ is
- A. $\{1, 2, 3\}$
B. $\{1, 2, 3, 6\}$
C. $\{2, 3, 4, 6\}$
D. $\{1, 6\}$
- 39.** If $P = \{\text{factors of } 12\}$ and $Q = \{\text{factors of } 18\}$, the number of elements in $P \cup Q$ is
- A. 14
B. 6
C. 4
D. 8
- 40.** In a market survey of 100 traders, 62 sell cloth, 48 sell shoes and 10 sell neither. The number of traders who sell both cloth and shoes is
- A. 20
B. 10
C. 30
D. 12

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Sets of factors and common factors.

Question 2

(a) Perfect squares.

Question 3

(a) Square roots.

Question 4

(a) Union and intersection of sets.

Question 5

(a) Venn diagrams and real-life problems.

Question 6

(a) Applied set problems.

Chapter 11: Ch3

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. $3.45 \times 100 =$
 - A. 34.5
 - B. 3,450
 - C. 345
 - D. 0.0345
2. $0.078 \times 1,000 =$
 - A. 78
 - B. 7.8
 - C. 780
 - D. 0.78
3. $6,250 \div 100 =$
 - A. 625
 - B. 6.25
 - C. 0.625
 - D. 62.5
4. $9.6 \div 10 =$
 - A. 96
 - B. 0.96
 - C. 0.096
 - D. 960
5. The benchmark fraction $\frac{1}{4}$ written as a decimal is
 - A. 0.25
 - B. 0.4
 - C. 0.14
 - D. 0.75
6. The decimal 0.2 written as a percentage is
 - A. 2%
 - B. 20%
 - C. 0.2%
 - D. 200%

- 7.** 25% written as a benchmark fraction is
- A. $\frac{1}{2}$
 - B. $\frac{1}{3}$
 - C. $\frac{1}{4}$
 - D. $\frac{3}{4}$
- 8.** Using the halving-and-doubling strategy, 24×35 can be written as $12 \times \square$. The missing number is
- A. 70
 - B. 17.5
 - C. 60
 - D. 84
- 9.** Using the halving-and-doubling strategy, the value of 24×35 is
- A. 420
 - B. 720
 - C. 980
 - D. 840
- 10.** The value of 56×25 , computed mentally, is
- A. 1,250
 - B. 1,500
 - C. 1,400
 - D. 1,450
- 11.** Recall: $12 \times 12 =$
- A. 124
 - B. 144
 - C. 132
 - D. 148
- 12.** Given that $7 \times 8 = 56$, the value of 70×80 is
- A. 5,600
 - B. 560
 - C. 56,000
 - D. 56
- 13.** A basket of oranges costs GH¢4.50. The cost of 20 such baskets, computed mentally, is
- A. GH¢45
 - B. GH¢95
 - C. GH¢90
 - D. GH¢80

14. $896,854 + 76,329 =$

- A. 963,183
- B. 973,283
- C. 972,183
- D. 973,183

15. $3,627.6 + 854.13 =$

- A. 4,471.73
- B. 4,481.73
- C. 4,481.63
- D. 4,381.73

16. $193.6 - 37.85 =$

- A. 155.75
- B. 166.25
- C. 156.75
- D. 155.25

17. $100,000 - 45,678 =$

- A. 55,322
- B. 54,422
- C. 64,322
- D. 54,322

18. $24,506 + 3,999 + 75,494 =$

- A. 104,999
- B. 102,999
- C. 103,999
- D. 103,099

19. $5.006 + 0.994 =$

- A. 5.996
- B. 6
- C. 6.4
- D. 5.1

20. $48,000 - 12,345 - 7,655 =$

- A. 30,000
- B. 27,000
- C. 29,000
- D. 28,000

- 21.** The sum of 10,001, 20,002 and 30,003 is
- A. 60,006
 - B. 60,060
 - C. 60,600
 - D. 66,000
- 22.** $657 \times 27 =$
- A. 17,379
 - B. 17,793
 - C. 17,739
 - D. 18,739
- 23.** $382 \times 856 =$
- A. 326,929
 - B. 326,992
 - C. 327,992
 - D. 336,992
- 24.** Using the distributive property, $325 \times 15 =$
- A. 4,875
 - B. 4,857
 - C. 4,785
 - D. 4,895
- 25.** $12,096 \div 24 =$
- A. 404
 - B. 54
 - C. 506
 - D. 504
- 26.** Which of the following numbers is divisible by 8?
- A. 3,422
 - B. 4,318
 - C. 4,312
 - D. 3,214
- 27.** Which of the following numbers is divisible by 9?
- A. 4,372
 - B. 5,346
 - C. 7,108
 - D. 2,438

- 28.** If a number is divisible by 6, it must also be divisible by
- A. 2 and 3
 - B. 4 and 3
 - C. 2 and 9
 - D. 3 and 5
- 29.** $48,000 \div 300 =$
- A. 16
 - B. 1,600
 - C. 160
 - D. 1,060
- 30.** The remainder when 10,000 is divided by 32 is
- A. 8
 - B. 16
 - C. 24
 - D. 0
- 31.** $234 \times 405 =$
- A. 94,700
 - B. 93,770
 - C. 94,670
 - D. 94,770
- 32.** Kofi bought 8 notebooks at GH¢12.00 each and Ama bought 12 pens at GH¢5.00 each. The total amount they spent altogether was
- A. GH¢156
 - B. GH¢146
 - C. GH¢166
 - D. GH¢96
- 33.** An amount of GH¢2,477.25 was shared equally among three children. Each child received
- A. GH¢825.25
 - B. GH¢725.75
 - C. GH¢825.75
 - D. GH¢826.75
- 34.** A pack of chocolate contained 250 bars. Adwoa took 90 bars and shared the rest equally among her four friends. Each friend received
- A. 62 bars
 - B. 40 bars
 - C. 50 bars
 - D. 35 bars

- 35.** Mrs Yaboi bought 25.25 metres of cloth for her five children. If they shared the material equally, each child received
- A. 5.5 m
 - B. 5.25 m
 - C. 5.005 m
 - D. 5.05 m
- 36.** A school approved GH¢2,940 for painting the school building and GH¢8,740 for buying new cadet uniforms. The total amount for the two projects was
- A. GH¢11,680
 - B. GH¢11,580
 - C. GH¢10,680
 - D. GH¢11,670
- 37.** From the same school budget, GH¢4,250 was approved for mending cracks on the netball pitch and GH¢9,990 for restocking the computer laboratory. The amount approved for the computer laboratory was more than that for the netball pitch by
- A. GH¢5,640
 - B. GH¢5,840
 - C. GH¢5,740
 - D. GH¢6,740
- 38.** A car travels at a constant speed of 72 km/h. The distance it travels in 4.5 hours is
- A. 314 km
 - B. 324 km
 - C. 334 km
 - D. 320 km
- 39.** 15.6 kg of rice is shared equally into 12 bags. The mass of rice in each bag is
- A. 1.2 kg
 - B. 1.25 kg
 - C. 1.4 kg
 - D. 1.3 kg
- 40.** Abena earns GH¢1,250 a month. Her total income in $2\frac{1}{2}$ years is
- A. GH¢37,500
 - B. GH¢30,000
 - C. GH¢37,050
 - D. GH¢36,250

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Powers of 10 and benchmark fractions.

Question 2

(a) Mental mathematics strategies.

Question 3

(a) Addition and subtraction.

Question 4

(a) Multiplication and division.

Question 5

(a) Divisibility tests.

Question 6

(a) Story problems.

Chapter 12: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. According to the laws of indices, $a^m \times a^n =$

- A. a^{mn}
- B. a^{m-n}
- C. a^{m+n}
- D. $(a^m)^n$

2. $a^m \div a^n =$

- A. a^{m-n}
- B. a^{m+n}
- C. $a^{m/n}$
- D. a^{n-m}

3. $(a^m)^n =$

- A. a^{m+n}
- B. $a^{m/n}$
- C. a^{m-n}
- D. a^{mn}

4. $(ab)^n =$

- A. ab^n
- B. $a^n b^n$
- C. $a^n + b^n$
- D. $a^n b$

5. For any non-zero number a , the value of a^0 is

- A. 1
- B. 0
- C. a
- D. undefined

6. Written with a positive index, $a^{-n} =$

- A. $-a^n$
- B. $1/a^n$
- C. a/n
- D. n/a

7. Which of the following states Law 3 of indices?

- A. $a^m \times a^n = a^{m+n}$
- B. $a^m \div a^n = a^{m-n}$
- C. $(a^m)^n = a^{mn}$
- D. $(ab)^n = a^n b^n$

8. The index form of $4 \times 4 \times 4 \times 4 \times 4$ is

- A. 4^5
- B. 5^4
- C. 4^4
- D. 20

9. $2^5 \times 2^3 =$

- A. 2^{15}
- B. 4^8
- C. 2^2
- D. 2^8

10. $3^7 \div 3^4 =$

- A. 3^{11}
- B. 1^3
- C. 3^3
- D. 3^{28}

11. $(5^2)^3 =$

- A. 5^5
- B. 15,625
- C. 3,125
- D. 25,625

12. $(2 \times 3)^4 =$

- A. 1,296
- B. 24
- C. 1,269
- D. 48

13. $5 \times 16^2 =$

- A. 1,208
- B. 5,120
- C. 1,280
- D. 12,800

14. Evaluate $(16/81)^{-3/4}$

- A. $8/27$
- B. $27/8$
- C. $-27/8$
- D. $3/2$

15. The fraction $27/32$ written in index form with prime bases is

- A. $3^3/2^4$
- B. $2^5/3^3$
- C. $3^3 \times 2^5$
- D. $3^3/2^5$

16. Simplify: $7^5 \div 7^2$

- A. 343
- B. 7^7
- C. 49
- D. 5

17. Simplify: $(3^2 \times 3^5) \div 3^4$

- A. 3^{11}
- B. 9
- C. 27
- D. 81

18. Simplify: $(2^3)^2 \times 2^4$

- A. 2^9
- B. 1,024
- C. 2^6
- D. 1,204

19. Evaluate: 5^{-2}

- A. -25
- B. $1/10$
- C. -10
- D. $1/25$

20. Simplify: $a^3 \times a^{-5}$

- A. $1/a^2$
- B. a^8
- C. a^2
- D. $1/a^8$

21. Solve: $25 = 5^{2x}$

- A. 5
- B. 2
- C. 1
- D. 25

22. Solve: $2^{x+2} = 16$

- A. 4
- B. 2
- C. 6
- D. 8

23. Solve: $2^5/2^3 = 2^{2x}$

- A. 2
- B. 4
- C. $\frac{1}{2}$
- D. 1

24. Solve: $1/27 = 3^x$

- A. -3
- B. 3
- C. -9
- D. 9

25. Solve: $3^{2x+1} = 27$

- A. 2
- B. 3
- C. 1
- D. 4

26. Solve: $4^x = 64$

- A. 2
- B. 3
- C. 4
- D. 16

27. If $2^x = 32$, then $2^{x+1} =$

- A. 33
- B. 16
- C. 128
- D. 64

- 28.** Solve: $9^x = 27$
- A. $3/2$
 - B. 3
 - C. 2
 - D. 1
- 29.** If $5^x = 1$, then $x =$
- A. 1
 - B. 5
 - C. 0
 - D. -1
- 30.** Solve: $10^x = 1,000,000$
- A. 5
 - B. 6
 - C. 7
 - D. 10
- 31.** Saratu traces her ancestors back 12 generations. The number of ancestors she had in the 12th generation is
- A. 2,048
 - B. 8,192
 - C. 24
 - D. 4,096
- 32.** The total number of Saratu's ancestors in the past 12 generations is
- A. 8,190
 - B. 4,096
 - C. 8,192
 - D. 4,094
- 33.** The number of ancestors in generation n is given by
- A. $2n$
 - B. n^2
 - C. 2^n
 - D. $2 + n$
- 34.** A bacterium divides into two every hour. Starting with one bacterium, the number of bacteria after 6 hours is
- A. 12
 - B. 64
 - C. 32
 - D. 62

- 35.** A sheet of paper is folded in half four times. The number of layers of paper formed is
- A. 8
 - B. 4
 - C. 12
 - D. 16
- 36.** The area of a square of side 3^4 cm is
- A. $6,561 \text{ cm}^2$
 - B. 81 cm^2
 - C. 324 cm^2
 - D. $2,187 \text{ cm}^2$
- 37.** The volume of a cube of side 2^3 cm is
- A. 24 cm^3
 - B. 64 cm^3
 - C. 512 cm^3
 - D. 216 cm^3
- 38.** Kofi's investment of GH¢50 doubles in value every year. Its value after 3 years is
- A. GH¢150
 - B. GH¢400
 - C. GH¢300
 - D. GH¢800
- 39.** A message is shared so that each person shares it with 3 new people. The number of new people who receive it in the 4th round is
- A. 27
 - B. 12
 - C. 64
 - D. 81
- 40.** The population of a town doubles every 10 years. If the population is now 5,000, it will be 40,000 after
- A. 30 years
 - B. 20 years
 - C. 40 years
 - D. 80 years

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) The laws of indices.

Question 2

(a) Evaluating powers.

Question 3

(a) Exponential equations.

Question 4

(a) Simplifying with the laws of indices.

Question 5

(a) Saratu's ancestors.

Question 6

(a) Real-life applications.

Chapter 13: Ch5

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The fraction $\frac{6}{10}$ expressed in its simplest form is
 - A. $\frac{6}{5}$
 - B. $\frac{2}{5}$
 - C. $\frac{3}{10}$
 - D. $\frac{3}{5}$
2. Which of the following fractions is equivalent to $\frac{2}{3}$?
 - A. $\frac{8}{12}$
 - B. $\frac{6}{8}$
 - C. $\frac{4}{5}$
 - D. $\frac{9}{12}$
3. The fraction $\frac{12}{5}$ written as a mixed number is
 - A. $2 \frac{1}{5}$
 - B. $3 \frac{2}{5}$
 - C. $2 \frac{2}{5}$
 - D. $2 \frac{3}{5}$
4. The mixed number $2 \frac{5}{9}$ written as an improper fraction is
 - A. $\frac{19}{9}$
 - B. $\frac{23}{9}$
 - C. $\frac{25}{9}$
 - D. $\frac{14}{9}$
5. $\frac{3}{4}$ of 60 is
 - A. 40
 - B. 20
 - C. 15
 - D. 45
6. Which of the following fractions is the greatest?
 - A. $\frac{5}{6}$
 - B. $\frac{3}{4}$
 - C. $\frac{2}{3}$
 - D. $\frac{7}{10}$

- 7.** The fraction $\frac{13}{8}$ written as a mixed number is
- A. $1 \frac{3}{8}$
 - B. $2 \frac{5}{8}$
 - C. $1 \frac{5}{8}$
 - D. $1 \frac{6}{8}$
- 8.** The shaded portion of a shape is 0.75. Written as a fraction in its simplest form, this is
- A. $\frac{7}{5}$
 - B. $\frac{3}{4}$
 - C. $\frac{75}{100}$
 - D. $\frac{4}{3}$
- 9.** $\frac{3}{4} + \frac{7}{8} =$
- A. $\frac{10}{12}$
 - B. $1 \frac{1}{8}$
 - C. $\frac{13}{16}$
 - D. $1 \frac{5}{8}$
- 10.** $\frac{4}{5} - \frac{1}{6} =$
- A. $\frac{19}{30}$
 - B. $\frac{3}{30}$
 - C. $\frac{19}{11}$
 - D. $\frac{5}{11}$
- 11.** $\frac{2}{3} \times \frac{3}{4} =$
- A. $\frac{6}{7}$
 - B. $\frac{5}{12}$
 - C. $\frac{1}{2}$
 - D. $\frac{8}{9}$
- 12.** $\frac{5}{8} \div 2 \frac{1}{2} =$
- A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
 - C. $2 \frac{1}{2}$
 - D. $\frac{25}{16}$
- 13.** $1 \frac{1}{2} + 2 \frac{3}{4} =$
- A. $3 \frac{4}{6}$
 - B. $3 \frac{1}{4}$
 - C. $4 \frac{1}{2}$
 - D. $4 \frac{1}{4}$

14. $3\frac{1}{2} - 1\frac{3}{4} =$

- A. $1\frac{3}{4}$
- B. $2\frac{1}{4}$
- C. $1\frac{1}{4}$
- D. $2\frac{3}{4}$

15. $\frac{2}{5} \times \frac{3}{7} =$

- A. $\frac{5}{12}$
- B. $\frac{6}{12}$
- C. $\frac{6}{35}$
- D. $\frac{5}{35}$

16. $\frac{3}{4} \div \frac{1}{8} =$

- A. $\frac{3}{32}$
- B. 6
- C. $\frac{1}{6}$
- D. 24

17. The reciprocal of $\frac{5}{9}$ is

- A. $-\frac{5}{9}$
- B. $\frac{5}{9}$
- C. $1\frac{5}{9}$
- D. $\frac{9}{5}$

18. $\frac{7}{12} + \frac{1}{4} =$

- A. $\frac{5}{6}$
- B. $\frac{8}{16}$
- C. $\frac{7}{48}$
- D. $\frac{2}{3}$

19. Using the order of operations, $\frac{3}{4} + \frac{5}{8} \times \frac{4}{5} - \frac{1}{6} =$

- A. $1\frac{5}{12}$
- B. $\frac{11}{12}$
- C. $1\frac{1}{12}$
- D. $\frac{13}{24}$

20. Using the order of operations, $\frac{3}{4} \div \frac{3}{8} + (\frac{4}{5} - \frac{1}{2}) =$

- A. $2\frac{1}{5}$
- B. $2\frac{3}{10}$
- C. $1\frac{7}{10}$
- D. $2\frac{1}{2}$

- 21.** Using the order of operations, $(\frac{3}{4} + \frac{5}{8}) \times \frac{4}{11} - \frac{1}{2} =$
- A. 0
 - B. $\frac{1}{2}$
 - C. 1
 - D. $\frac{1}{4}$
- 22.** $\frac{2}{3}$ of $(\frac{1}{2} + \frac{1}{6}) =$
- A. $\frac{2}{9}$
 - B. $\frac{1}{3}$
 - C. $\frac{2}{3}$
 - D. $\frac{4}{9}$
- 23.** $(\frac{5}{6} - \frac{1}{3}) \div \frac{1}{2} =$
- A. $\frac{1}{4}$
 - B. 2
 - C. 1
 - D. $\frac{3}{4}$
- 24.** $\frac{1}{2} + \frac{1}{4} \times \frac{2}{3} =$
- A. $\frac{1}{2}$
 - B. $\frac{2}{3}$
 - C. $\frac{1}{3}$
 - D. $\frac{5}{12}$
- 25.** $(2\frac{1}{2} - \frac{1}{2}) \times \frac{3}{4} =$
- A. $1\frac{1}{2}$
 - B. $\frac{3}{4}$
 - C. 1
 - D. $2\frac{1}{4}$
- 26.** $1\frac{1}{4} \div \frac{1}{2} + \frac{1}{4} =$
- A. $1\frac{1}{2}$
 - B. 2
 - C. $2\frac{3}{4}$
 - D. $2\frac{1}{2}$
- 27.** Faako answered 42 out of 60 questions correctly. The percentage of her answers that were correct is
- A. 60%
 - B. 70%
 - C. 42%
 - D. 75%

- 28.** In a school, $\frac{2}{3}$ of the students eat from the school feeding programme and $\frac{1}{4}$ bring packed lunch. The fraction of the students who go home for lunch is
- A. $\frac{1}{12}$
 - B. $\frac{5}{12}$
 - C. $\frac{1}{7}$
 - D. $\frac{11}{12}$
- 29.** Esi's orange drink is made with $\frac{2}{7}$ orange squash and Fusena's with $\frac{1}{4}$ orange squash. The drink that tastes stronger of orange is
- A. both taste equally strong
 - B. neither, since both are less than a half
 - C. Fusena's, because $\frac{1}{4}$ is greater than $\frac{2}{7}$
 - D. Esi's, because $\frac{2}{7}$ is greater than $\frac{1}{4}$
- 30.** The perimeter of a rectangle whose sides measure $1\frac{1}{3}$ cm and $3\frac{3}{4}$ cm is
- A. 5 cm
 - B. $9\frac{1}{6}$ cm
 - C. $10\frac{1}{6}$ cm
 - D. $20\frac{1}{3}$ cm
- 31.** The area of a rectangle whose sides measure $1\frac{1}{3}$ cm and $3\frac{3}{4}$ cm is
- A. 4 cm^2
 - B. 5 cm^2
 - C. 6 cm^2
 - D. $4\frac{1}{2}\text{ cm}^2$
- 32.** Ama had GH¢240.00 and spent $\frac{2}{3}$ of it. The amount she spent was
- A. GH¢120.00
 - B. GH¢80.00
 - C. GH¢160.00
 - D. GH¢180.00
- 33.** A rope $12\frac{1}{2}$ m long is cut into 5 equal pieces. The length of each piece is
- A. $2\frac{1}{2}$ m
 - B. $2\frac{1}{4}$ m
 - C. 2 m
 - D. 3 m
- 34.** A tank is $\frac{3}{8}$ full of water. The fraction of the tank that is empty is
- A. $\frac{3}{8}$
 - B. $\frac{1}{8}$
 - C. $\frac{8}{5}$
 - D. $\frac{5}{8}$

- 35.** $\frac{3}{5}$ of the 45 students in a class are girls. The number of boys in the class is
- A. 27
 - B. 15
 - C. 30
 - D. 18
- 36.** Written as a decimal and rounded to two decimal places, $\frac{2}{3}$ is
- A. 0.67
 - B. 0.66
 - C. 0.6
 - D. 0.7
- 37.** Kofi read $\frac{1}{4}$ of a 240-page book on Monday and $\frac{1}{3}$ of it on Tuesday. The total number of pages he read on the two days is
- A. 100
 - B. 120
 - C. 140
 - D. 160
- 38.** Kofi read $\frac{1}{4}$ of a 240-page book on Monday and $\frac{1}{3}$ of it on Tuesday. The number of pages left unread is
- A. 140
 - B. 100
 - C. 80
 - D. 60
- 39.** A farmer uses $\frac{1}{5}$ of his land for maize and $\frac{1}{4}$ of the remainder for vegetables. The fraction of the whole land used for vegetables is
- A. $\frac{1}{9}$
 - B. $\frac{1}{20}$
 - C. $\frac{1}{4}$
 - D. $\frac{1}{5}$
- 40.** A drink is made by mixing 2 cups of orange squash with 5 cups of water. The fraction of the drink that is squash is
- A. $\frac{2}{7}$
 - B. $\frac{2}{5}$
 - C. $\frac{5}{7}$
 - D. $\frac{1}{2}$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Review of fractions.

Question 2

(a) Operations on fractions.

Question 3

(a) Order of operations (BODMAS).

Question 4

(a) Fractions of quantities.

Question 5

(a) Measurement with fractions.

Question 6

(a) Integrated problems.

Chapter 14: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. 4 km written in metres is
 - A. 400 m
 - B. 40 m
 - C. 40,000 m
 - D. 4,000 m
2. 250 cm written in metres is
 - A. 2.5 m
 - B. 25 m
 - C. 0.25 m
 - D. 2,500 m
3. 3.2 litres written in millilitres is
 - A. 320 mL
 - B. 32 mL
 - C. 3,200 mL
 - D. 32,000 mL
4. 1 hour 15 minutes written in minutes is
 - A. 115 minutes
 - B. 75 minutes
 - C. 65 minutes
 - D. 90 minutes
5. Using ratio reasoning, since $1 \text{ m} = 100 \text{ cm}$, a measurement in metres is converted to centimetres by multiplying by
 - A. $1/100$
 - B. 10
 - C. 1,000
 - D. 100
6. Agbo walks 4 km to school every day and uses 60 minutes. His speed is
 - A. 4 km/h
 - B. 240 km/h
 - C. 0.067 km/h
 - D. 1 km/h

- 7.** Rukiya covers 4,200 m in 45 minutes. Her speed is
- A. 4.2 km/h
 - B. 4.5 km/h
 - C. 5.6 km/h
 - D. 9.3 km/h
- 8.** Agbo walks 4 km in 60 minutes and Rukiya covers 4,200 m in 45 minutes. The faster walker is
- A. Agbo
 - B. Rukiya
 - C. both are equally fast
 - D. it cannot be determined
- 9.** If it takes 7 hours to mow 4 lawns, then at the same rate, the number of lawns that could be mowed in 35 hours is
- A. 12
 - B. 28
 - C. 35
 - D. 20
- 10.** The rate of mowing, if 4 lawns are mowed in 7 hours, is
- A. $\frac{4}{7}$ of a lawn per hour
 - B. $\frac{7}{4}$ of a lawn per hour
 - C. 28 lawns per hour
 - D. 4 lawns per hour
- 11.** Salamatu burns 756 calories while drumming for 3 hours. The number of calories she burns per hour is
- A. 2,268
 - B. 225
 - C. 252
 - D. 352
- 12.** Six exercise books cost GH¢42.00. The cost of one exercise book is
- A. GH¢6.00
 - B. GH¢7.00
 - C. GH¢8.00
 - D. GH¢252.00
- 13.** A car travels 240 km in 3 hours. Its constant speed is
- A. 720 km/h
 - B. 60 km/h
 - C. 90 km/h
 - D. 80 km/h

- 14.** A car travels at a constant speed of 80 km/h. The time it takes to travel 400 km is
- A. 5 hours
 - B. 4 hours
 - C. 320 hours
 - D. 6 hours
- 15.** Three kilograms of rice cost GH¢27.00. The cost of 5 kg at the same rate is
- A. GH¢36.00
 - B. GH¢40.50
 - C. GH¢45.00
 - D. GH¢54.00
- 16.** A machine produces 1,200 bottles in 8 hours. The number of bottles produced per hour is
- A. 120
 - B. 150
 - C. 160
 - D. 9,600
- 17.** A farmer uses 4 bags of fertilizer on 5 acres of land. The number of bags needed for 20 acres is
- A. 8
 - B. 12
 - C. 20
 - D. 16
- 18.** Kofi runs 1,500 m in 5 minutes. His time per kilometre is
- A. 3 $\frac{1}{3}$ minutes
 - B. 3 minutes
 - C. 4 minutes
 - D. 5 minutes
- 19.** On a distance–time graph, a horizontal line means the object is
- A. moving with increasing speed
 - B. moving at a constant speed
 - C. at rest
 - D. moving backwards
- 20.** The gradient of a distance–time graph represents
- A. distance
 - B. speed
 - C. time
 - D. acceleration

- 21.** A trader travels from Buduata to Adawso, a distance of 20 miles. After 60 minutes she stops at Assin, which is 8 miles from Buduata. Her speed during the first hour is
- A. 8 miles/h
 - B. 20 miles/h
 - C. 12 miles/h
 - D. 0.13 miles/h
- 22.** In the same journey, the trader covers the remaining 12 miles to Adawso in 24 minutes. Her speed during this part of the journey is
- A. 12 miles/h
 - B. 24 miles/h
 - C. 0.5 miles/h
 - D. 30 miles/h
- 23.** On a distance–time graph, a steep straight line shows that the object is
- A. at rest
 - B. slowing down
 - C. moving at a high constant speed
 - D. travelling a decreasing distance
- 24.** A cyclist covers 15 km in 45 minutes. His speed in km/h is
- A. 15 km/h
 - B. 20 km/h
 - C. 45 km/h
 - D. 11.25 km/h
- 25.** A bus travels at 60 km/h for $2\frac{1}{2}$ hours. The distance it covers is
- A. 150 km
 - B. 120 km
 - C. 24 km
 - D. 160 km
- 26.** From a travel graph, Ama leaves home at 7:00 and walks 4 km to school, arriving at 8:00. Her average speed is
- A. 1 km/h
 - B. 8 km/h
 - C. 4 km/h
 - D. 2 km/h
- 27.** Two quantities are proportional if
- A. their sum is constant
 - B. their ratio is constant
 - C. their product is constant
 - D. their difference changes

- 28.** Which of the following tables shows a proportional relationship?
- A. x : 1, 2, 3 and y : 5, 10, 16
 - B. x : 1, 2, 3 and y : 5, 10, 15
 - C. x : 1, 2, 3 and y : 5, 11, 17
 - D. x : 1, 2, 3 and y : 5, 9, 13
- 29.** A straight-line graph which does NOT pass through the origin represents
- A. a proportional relationship
 - B. a unit rate of zero
 - C. a non-proportional relationship
 - D. no relationship
- 30.** Two pens cost GH¢6.00 and five pens cost GH¢15.00. This shows
- A. a non-proportional relationship
 - B. a constant sum
 - C. a proportional relationship
 - D. a decreasing pattern
- 31.** The equation $y = 3x$ represents a proportional relationship because
- A. it has a constant term
 - B. its graph is a straight line through the origin
 - C. y increases by 3 each time
 - D. x is always positive
- 32.** Which of the following equations represents a proportional relationship?
- A. $y = 2x + 1$
 - B. $y = x - 3$
 - C. $y = 5x$
 - D. $y = 4 - x$
- 33.** The ratios 3 : 4 and 9 : 12 are equivalent, so the two quantities are
- A. proportional
 - B. not proportional
 - C. equal
 - D. undefined
- 34.** An ant travels $\frac{9}{8}$ of an inch in 45 seconds. The constant of proportionality, in inches per second, is
- A. $\frac{9}{8}$
 - B. 45
 - C. $\frac{1}{8}$
 - D. $\frac{1}{40}$

- 35.** The same ant travels $27/8$ of an inch in 2 minutes 15 seconds. The constant of proportionality is
- A. $1/20$
 - B. $1/40$
 - C. $27/8$
 - D. 135
- 36.** A table shows distances of 40 km in 2 h, 60 km in 3 h and 100 km in 5 h. The constant of proportionality is
- A. 20 km/h
 - B. 80 km/h
 - C. 200 km/h
 - D. 5 km/h
- 37.** In the proportional relationship $y = kx$, $y = 36$ when $x = 4$. The value of k is
- A. 144
 - B. 32
 - C. 9
 - D. 40
- 38.** The constant of proportionality is also known as the
- A. constant sum
 - B. unit rate
 - C. y-intercept
 - D. total amount
- 39.** Five kilograms of oranges cost GH¢60.00. The constant of proportionality (cost per kilogram) is
- A. GH¢300.00
 - B. GH¢5.00
 - C. GH¢60.00
 - D. GH¢12.00
- 40.** If $y = kx$ with $k = 12$, the value of y when $x = 7$ is
- A. 84
 - B. 19
 - C. 5
 - D. 12

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Unit conversion by ratio reasoning.

Question 2

(a) Unit rate problems.

Question 3

(a) Speed and distance–time graphs.

Question 4

(a) Proportional relationships.

Question 5

(a) Constant of proportionality.

Question 6

(a) Integrated problems.

Chapter 15: Ch7

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The gradient of a straight line is a measure of its
 - A. length
 - B. steepness
 - C. smoothness
 - D. distance from the origin
2. The formula for the gradient of the line through (x_1, y_1) and (x_2, y_2) is
 - A. $(x_2 - x_1)/(y_2 - y_1)$
 - B. $(y_2 + y_1)/(x_2 + x_1)$
 - C. $(x_2 - y_2)/(x_1 - y_1)$
 - D. $(y_2 - y_1)/(x_2 - x_1)$
3. A line which rises from left to right has a gradient which is
 - A. positive
 - B. negative
 - C. zero
 - D. undefined
4. The gradient of a horizontal line is
 - A. 1
 - B. undefined
 - C. 0
 - D. -1
5. In the equation $y = mx + c$, the letter m represents the
 - A. y-intercept
 - B. gradient
 - C. x-coordinate
 - D. origin
6. In the equation $y = mx + c$, the letter c represents the
 - A. gradient
 - B. x-intercept
 - C. constant gradient
 - D. y-intercept

- 7.** The gradient of the line passing through A(1, 1) and B(7, 2) is
- A. $1/6$
 - B. 6
 - C. $1/5$
 - D. $-1/6$
- 8.** The gradient of the line passing through P(-2, 4) and Q(3, 5) is
- A. 5
 - B. $1/3$
 - C. $1/5$
 - D. $-1/5$
- 9.** The gradient of the line passing through C(3, -2) and D(-3, 4) is
- A. 1
 - B. -1
 - C. 0
 - D. 6
- 10.** The gradient of the line passing through (0, 0) and (4, 10) is
- A. $2/5$
 - B. 40
 - C. 4
 - D. $5/2$
- 11.** The gradient of the line passing through (2, 5) and (2, 9) is
- A. undefined
 - B. 0
 - C. 1
 - D. 4
- 12.** A line passes through (1, 3) and has gradient 2. Which of the following points also lies on the line?
- A. (2, 6)
 - B. (3, 6)
 - C. (4, 9)
 - D. (5, 10)
- 13.** The gradient of the line $y = 5x + 13$ is
- A. 13
 - B. 5
 - C. 18
 - D. -5

- 14.** The gradient of the line $2x - 8y + 3 = 0$ is
- A. 2
 - B. -8
 - C. 3
 - D. $1/4$
- 15.** The gradient and the y-intercept of the line $y = -3x + 12$ are respectively
- A. -3 and 12
 - B. 3 and 12
 - C. -3 and -12
 - D. 12 and -3
- 16.** The equation of the line with gradient 2 and y-intercept -3 is
- A. $y = -3x + 2$
 - B. $y = 2x + 3$
 - C. $y = 2x - 3$
 - D. $y = 3x - 2$
- 17.** For the line $y = 2x - 3$, the value of y when $x = 4$ is
- A. 8
 - B. 5
 - C. 11
 - D. 2
- 18.** The equation of the line with gradient $1/2$ and y-intercept 4 is
- A. $y = 4x + 1/2$
 - B. $y = (1/2)x - 4$
 - C. $y = 2x + 4$
 - D. $y = (1/2)x + 4$
- 19.** The equation of the line with gradient $-5/2$ and y-intercept $7/2$ is
- A. $y = (-5/2)x + 7/2$
 - B. $y = (5/2)x + 7/2$
 - C. $y = (-5/2)x - 7/2$
 - D. $y = (7/2)x - 5/2$
- 20.** Using the point-slope form, the equation of the line with gradient $2/3$ passing through $(3, -1)$ is
- A. $y = (2/3)x + 3$
 - B. $y = (3/2)x - 3$
 - C. $y = (2/3)x - 3$
 - D. $y = (-2/3)x + 3$

- 21.** The equation of the line passing through (5, 4) and (−10, −2) is
- A. $y = (2/5)x - 2$
 - B. $y = (2/5)x + 2$
 - C. $y = (5/2)x + 2$
 - D. $y = (-2/5)x + 2$
- 22.** Written in the form $y = mx + a$, the equation $5x + 4y - 3 = 0$ becomes
- A. $y = (5/4)x + 3/4$
 - B. $y = (-5/4)x - 3/4$
 - C. $y = (4/5)x + 3$
 - D. $y = (-5/4)x + 3/4$
- 23.** The gradient and the y-intercept of the line $5x + 4y - 3 = 0$ are respectively
- A. $-5/4$ and $3/4$
 - B. $5/4$ and $3/4$
 - C. $-5/4$ and $-3/4$
 - D. $4/5$ and 3
- 24.** The equation of a straight line which passes through the origin with gradient 4 is
- A. $y = x + 4$
 - B. $y = 4$
 - C. $y = 4x$
 - D. $y = 4 - x$
- 25.** The cost of meat in a market is given by $y = 20x$, where y is the cost in cedis and x the weight in kg. The cost of 3.5 kg of meat is
- A. GH¢70
 - B. GH¢35
 - C. GH¢80
 - D. GH¢23.5
- 26.** Using the relation $y = 20x$, the weight of meat that can be bought with GH¢80 is
- A. 1,600 kg
 - B. 4 kg
 - C. 16 kg
 - D. 0.25 kg
- 27.** The diameter of a Wawa tree is modelled by $y = (2/5)x + 10$, where y is the diameter in inches after x years. The diameter after 20 years is
- A. 8 inches
 - B. 14 inches
 - C. 18 inches
 - D. 50 inches

- 28.** The diameter of the Wawa tree, using the relation $y = (2/5)x + 10$, in 100 years' time will be
- A. 40 inches
 - B. 60 inches
 - C. 110 inches
 - D. 50 inches
- 29.** For the relation $y = (2/5)x + 10$, the value of y when $x = 0$ is
- A. 10
 - B. 0
 - C. $2/5$
 - D. 2
- 30.** In the table of values for $y = 3x + 1$, the value of y when $x = 4$ is
- A. 12
 - B. 13
 - C. 7
 - D. 11
- 31.** The table x : 0, 1, 2, 3 and y : 2, 5, 8, 11 represents the relation
- A. $y = 2x + 3$
 - B. $y = x + 2$
 - C. $y = 3x + 2$
 - D. $y = 3x - 2$
- 32.** The gradient of the line represented by the table x : 0, 1, 2, 3 and y : 2, 5, 8, 11 is
- A. 2
 - B. 5
 - C. 1
 - D. 3
- 33.** The distance you walk is modelled by $d = (1/3)h$, where d is the distance in km and h the number of hours walked. The distance walked after 6 hours is
- A. 2 km
 - B. 18 km
 - C. 3 km
 - D. $1/2$ km
- 34.** Nhyira charges GH¢50 for every hour she spends painting a portrait. Her charge for a portrait which takes 3 hours is
- A. GH¢53
 - B. GH¢100
 - C. GH¢150
 - D. GH¢300

- 35.** At the same rate, if Nhyira charged GH¢175 for a portrait, the number of hours she used is
- A. 3 hours
 - B. $3\frac{1}{2}$ hours
 - C. 4 hours
 - D. 125 hours
- 36.** The number of hours Nhyira will require to paint a portrait costing GH¢300 is
- A. 3 hours
 - B. 5 hours
 - C. $4\frac{1}{2}$ hours
 - D. 6 hours
- 37.** A taxi charges a flat rate of GH¢5 plus GH¢2 for every kilometre. The equation of the cost y cedis for a journey of x km is
- A. $y = 2x + 5$
 - B. $y = 5x + 2$
 - C. $y = 7x$
 - D. $y = 2x - 5$
- 38.** Using the taxi fares above, the cost of a 10 km journey is
- A. GH¢20
 - B. GH¢50
 - C. GH¢25
 - D. GH¢15
- 39.** A worker saves GH¢30 every month, starting with GH¢100. The equation of her savings y cedis after x months is
- A. $y = 100x + 30$
 - B. $y = 30x + 100$
 - C. $y = 130x$
 - D. $y = 30x - 100$
- 40.** Using the savings equation above, the worker's savings after 6 months will be
- A. GH¢180
 - B. GH¢310
 - C. GH¢210
 - D. GH¢280

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) The gradient of a straight line.

Question 2

- (a) Equations of straight lines.

Question 3

- (a) Point-slope form and rearranging equations.

Question 4

- (a) Tables of values and graphs.

Question 5

- (a) Real-life linear graphs.

Question 6

- (a) The Wawa tree.

Chapter 16: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. When $6(x + 3)$ is expanded, the result is
 - A. $6x + 3$
 - B. $9x$
 - C. $6x + 18$
 - D. $6x^2 + 3$
2. When $-5x(3x + 4)$ is expanded, the result is
 - A. $-15x^2 + 20x$
 - B. $-15x^2 - 20x$
 - C. $-8x^2 - 9x$
 - D. $15x^2 - 20x$
3. The expression $3(x + 4) - 2(x - 5)$, when simplified, is
 - A. $x - 2$
 - B. $x + 2$
 - C. $5x + 2$
 - D. $x + 22$
4. The expression $2(6 - 5x) - 3(2 + 2x) - 4(3x - 1)$, when simplified, is
 - A. $10 - 28x$
 - B. $10 + 28x$
 - C. $2 - 28x$
 - D. $22 - 4x$
5. The expression $8 - (4 - a) - (6 - a)$, when simplified, is
 - A. $2 - 2a$
 - B. $2a - 2$
 - C. $18 - 2a$
 - D. $2a + 2$
6. The expression $(m + o - g) - (m - o + g)$, when simplified, is
 - A. $2o - 2g$
 - B. $2m - 2g$
 - C. $-2o + 2g$
 - D. $2m + 2o$

- 7.** When $4a(2a - 5)$ is expanded, the result is
- A. $8a - 20a$
 - B. $8a^2 - 5$
 - C. $6a^2 - 20a$
 - D. $8a^2 - 20a$
- 8.** When $-3(2 - 4y)$ is expanded, the result is
- A. $-6 - 12y$
 - B. $6 - 12y$
 - C. $-6 + 12y$
 - D. $-6 + 12y^2$
- 9.** $(a + 2)(a + 3) =$
- A. $a^2 + 6$
 - B. $a^2 + 5a + 6$
 - C. $a^2 + 6a + 5$
 - D. $2a^2 + 5a + 6$
- 10.** $(2x + y)(2x - y) =$
- A. $4x^2 - y^2$
 - B. $4x^2 + y^2$
 - C. $2x^2 - y^2$
 - D. $4x^2 - 2xy - y^2$
- 11.** $(3x - 2y)(3x + 2y) =$
- A. $9x^2 + 4y^2$
 - B. $9x^2 - 12xy + 4y^2$
 - C. $6x^2 - 4y^2$
 - D. $9x^2 - 4y^2$
- 12.** $(2x + 3)^2 =$
- A. $4x^2 + 9$
 - B. $4x^2 + 6x + 9$
 - C. $4x^2 + 12x + 9$
 - D. $2x^2 + 12x + 9$
- 13.** $(a + 2)^2 =$
- A. $a^2 + 4$
 - B. $a^2 + 4a + 4$
 - C. $a^2 + 2a + 4$
 - D. $2a^2 + 4$

14. $(x - 2)^2 =$

- A. $x^2 - 4x + 4$
- B. $x^2 - 4$
- C. $x^2 + 4$
- D. $x^2 - 2x + 4$

15. $(x + 5)(x - 3) =$

- A. $x^2 - 15$
- B. $x^2 + 8x - 15$
- C. $x^2 - 2x + 15$
- D. $x^2 + 2x - 15$

16. $(p + q)(p - q) =$

- A. $p^2 + q^2$
- B. $p^2 - q^2$
- C. $p^2 - 2pq + q^2$
- D. $2p - 2q$

17. The expression $(a - b)/3 + 3a/2$, when simplified, is

- A. $(11a - 2b)/6$
- B. $(4a - b)/5$
- C. $(11a - 2b)/5$
- D. $(2a - 2b + 9a)/5$

18. The expression $2a/3 - (a - b)/2$, when simplified, is

- A. $(a - 3b)/6$
- B. $(4a - a - b)/6$
- C. $(a + 3b)/6$
- D. $(3a + 2b)/6$

19. The expression $5/6r - 3/4r$, when simplified, is

- A. $2/2r$
- B. $1/12r^2$
- C. $1/2r$
- D. $1/12r$

20. The expression $(2x - 1)/3 - (x + 3)/2$, when simplified, is

- A. $(x + 2)/6$
- B. $(x - 11)/6$
- C. $(x - 11)/5$
- D. $(3x - 4)/6$

- 21.** The expression $(3a + 5b)/4 + (a + b)/8$, when simplified, is
- A. $(7a + 11b)/8$
 - B. $(4a + 6b)/8$
 - C. $(7a + 11b)/12$
 - D. $(6a + 10b)/8$
- 22.** The expression $x/3 + x/6$, when simplified, is
- A. $2x/9$
 - B. $x/9$
 - C. $x/2$
 - D. $2x/6^2$
- 23.** The product of $3a/4$ and $8a/9$ is
- A. $11a/13$
 - B. $2a^2/3$
 - C. $3a^2/2$
 - D. $2a/3$
- 24.** $10x/3$ divided by $5x/9$ is
- A. $50x^2/27$
 - B. $3/2$
 - C. $2/3$
 - D. 6
- 25.** If $x = 2$, the value of $3/(x + 1) - 2/(x - 1)$ is
- A. -1
 - B. 1
 - C. 5
 - D. -5
- 26.** If $x = 2$, the value of $1/(x - 1) + 2/(x + 1)$ is
- A. 3
 - B. 1
 - C. $2/3$
 - D. $5/3$
- 27.** If $x = 2$, $y = -2$, $z = 3$ and $a = 1$, the value of $5x + 7z^2 - 4a + 3y^2$ is
- A. 63
 - B. 81
 - C. 45
 - D. 87

- 28.** If $x = 2$ and $y = -2$, the value of $x^2 + y^2$ is
- A. 0
 - B. 4
 - C. 8
 - D. -8
- 29.** If $x = 2$, $y = -2$ and $z = 3$, the value of $2x + 3y - z$ is
- A. -5
 - B. 5
 - C. 1
 - D. -1
- 30.** If $a = 1$ and $b = -1$, the value of $a^2 - b^2$ is
- A. 2
 - B. -2
 - C. 1
 - D. 0
- 31.** If $x = 2$ and $y = -2$, the value of $(12xy/7) \times (14x/20)$ is
- A. $48/5$
 - B. $24/5$
 - C. $-48/5$
 - D. $-24/5$
- 32.** If $x = 2$, the value of $3x^2 - 5x + 4$ is
- A. 6
 - B. 10
 - C. 2
 - D. 16
- 33.** The expression $3ax + 6ay$ factorised completely is
- A. $3(ax + 2y)$
 - B. $6a(x + y)$
 - C. $3a(x + 6y)$
 - D. $3a(x + 2y)$
- 34.** The expression $54 - 81x$ factorised completely is
- A. $9(6 - 9x)$
 - B. $27(2 - 3x)$
 - C. $27(3 - 2x)$
 - D. $54(1 - 81x)$

35. The expression $100x - 25x^2$ factorised completely is

- A. $25(4x - x^2)$
- B. $5x(20 - 5x)$
- C. $25x(4 - x)$
- D. $25x(x - 4)$

36. The expression $x^2 - 25$ factorised is

- A. $(x - 5)^2$
- B. $(x - 5)(x + 5)$
- C. $(x - 25)(x + 1)$
- D. $x(x - 25)$

37. The expression $x^2 + 5x + 6$ factorised is

- A. $(x + 2)(x + 3)$
- B. $(x + 1)(x + 6)$
- C. $(x + 5)(x + 1)$
- D. $(x - 2)(x - 3)$

38. The expression $4x^2 - 9y^2$ factorised is

- A. $(2x - 3y)^2$
- B. $(4x + 9y)(x - y)$
- C. $(2x - 3y)(2x + 3y)$
- D. $4x(x - 9y)$

39. The expression $ax - ay - xy + y^2$ factorised by grouping is

- A. $(x + y)(a - y)$
- B. $(x - y)(a + y)$
- C. $(x - a)(y - x)$
- D. $(x - y)(a - y)$

40. The expression $3x^2 + 2xy - 12xz - 8yz$ factorised by grouping is

- A. $(3x + 2y)(x + 4z)$
- B. $(3x + 2y)(x - 4z)$
- C. $(3x - 2y)(x - 4z)$
- D. $(3x + 8y)(x - z)$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Expanding brackets.

Question 2

(a) Multiplication of binomial expressions.

Question 3

(a) Operations on algebraic fractions.

Question 4

(a) Substitution and evaluation.

Question 5

(a) Factorisation.

Question 6

(a) Integrated algebra.

Chapter 17: Ch9

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A whole number is less than 17. If the number is x , the inequality representing the statement is
 - A. $x < 17$
 - B. $x > 17$
 - C. $x \leq 17$
 - D. $x \geq 17$
2. Eight less than the product of -3 and a number x is greater than -26 . The inequality representing the statement is
 - A. $-3x + 8 > -26$
 - B. $-3(x - 8) > -26$
 - C. $-3x - 8 > -26$
 - D. $8 - 3x < -26$
3. Kwaakye's profit for March was GH¢32.00, which was at least GH¢12.00 less than his profit for February. If his February profit is p cedis, the inequality representing the situation is
 - A. $32 \leq p - 12$
 - B. $32 \geq p - 12$
 - C. $p \geq 32 - 12$
 - D. $32 > p + 12$
4. Ama is older than 12 years. If her age is a years, the inequality representing the statement is
 - A. $a < 12$
 - B. $a = 12$
 - C. $a \geq 12$
 - D. $a > 12$
5. The total cost of x books at GH¢5.00 each does not exceed GH¢100.00. The inequality representing the statement is
 - A. $5x \leq 100$
 - B. $5x \geq 100$
 - C. $5x < 100$
 - D. $x + 5 \leq 100$

- 6.** The statement “ x is at most 9” means
- A. $x > 9$
 - B. $x < 9$
 - C. $x \leq 9$
 - D. $x \geq 9$
- 7.** A lift can carry at most 600 kg. If x bags of rice, each weighing 50 kg, are put into the lift, the inequality representing the statement is
- A. $50x \geq 600$
 - B. $50x \leq 600$
 - C. $x \leq 600$
 - D. $600x \leq 50$
- 8.** Which of the following word statements matches the inequality $2x + 5 > 15$?
- A. Twice the sum of a number and 5 is greater than 15
 - B. Five times a number, plus 2, is greater than 15
 - C. The sum of twice a number and 15 is greater than 5
 - D. Five more than twice a number is greater than 15
- 9.** Solve the inequality $x - 5 > 1$
- A. $x > 6$
 - B. $x > 4$
 - C. $x < 6$
 - D. $x > -4$
- 10.** Solve the inequality $x - 3 \geq 2$
- A. $x \geq 1$
 - B. $x \geq -1$
 - C. $x \leq 5$
 - D. $x \geq 5$
- 11.** Solve the inequality $13 - x < 12$
- A. $x < 1$
 - B. $x > 25$
 - C. $x > 1$
 - D. $x < 25$
- 12.** Solve the inequality $3x \leq 8 + x$
- A. $x \leq 8$
 - B. $x \leq 4$
 - C. $x \geq 4$
 - D. $x \leq 2$

13. Solve the inequality $x + 7 \leq 15$

- A. $x \leq 8$
- B. $x \geq 8$
- C. $x \leq 22$
- D. $x < 8$

14. Solve the inequality $2x - 13 > 29$

- A. $x > 8$
- B. $x > 16$
- C. $x > 42$
- D. $x > 21$

15. Solve the inequality $4x + 3 \geq 23$

- A. $x \geq 4$
- B. $x \geq 5$
- C. $x \leq 5$
- D. $x \geq 6$

16. Solve the inequality $-2x < 10$

- A. $x > -5$
- B. $x < -5$
- C. $x > 5$
- D. $x < 5$

17. Solve the inequality $7 - 2x \geq 1$

- A. $x \geq 3$
- B. $x \leq -3$
- C. $x \leq 3$
- D. $x \geq -3$

18. Solve the inequality $x/4 > 3$

- A. $x > 3/4$
- B. $x > 12$
- C. $x < 12$
- D. $x > 7$

19. Solve the inequality $5x - 9 > -5$

- A. $x > -4/5$
- B. $x > 14/5$
- C. $x < 4/5$
- D. $x > 4/5$

20. Solve the inequality $6 - x \leq 2$

- A. $x \geq 4$
- B. $x \leq 4$
- C. $x \geq -4$
- D. $x \leq 8$

21. Solve the inequality $x + 4 \leq 3x - 16$

- A. $x \leq 10$
- B. $x \geq 6$
- C. $x \geq 10$
- D. $x \leq 6$

22. The smallest whole number which satisfies $2x + 1 > 9$ is

- A. 4
- B. 5
- C. 9
- D. 10

23. If $x < 4$ and x is a whole number, the solution set is

- A. $\{1, 2, 3\}$
- B. $\{0, 1, 2, 3, 4\}$
- C. $\{4, 3, 2, 1\}$
- D. $\{0, 1, 2, 3\}$

24. If $2x > 24$ and x is a whole number, then

- A. $x > 12$
- B. $x < 12$
- C. $x > 48$
- D. $x \geq 12$

25. If $x \geq 5$ and x is a whole number less than 9, the solution set is

- A. $\{5, 6, 7\}$
- B. $\{5, 6, 7, 8\}$
- C. $\{6, 7, 8, 9\}$
- D. $\{5, 6, 7, 8, 9\}$

26. The solution set of $x + 2 \leq 5$, where x is a whole number, is

- A. $\{0, 1, 2\}$
- B. $\{1, 2, 3\}$
- C. $\{0, 1, 2, 3\}$
- D. $\{3, 2, 1, 0, -1\}$

- 27.** If $3x < 15$ and x is a positive whole number, the solution set is
- A. $\{1, 2, 3, 4\}$
 - B. $\{0, 1, 2, 3, 4\}$
 - C. $\{1, 2, 3, 4, 5\}$
 - D. $\{5, 6, 7, \dots\}$
- 28.** The inequality satisfied by all the numbers in the set $\{6, 7, 8, \dots\}$ is
- A. $x > 8$
 - B. $x \leq 6$
 - C. $x < 6$
 - D. $x \geq 6$
- 29.** If $x - 1 > 2$ and x is a whole number at most 6, the solution set is
- A. $\{3, 4, 5, 6\}$
 - B. $\{4, 5, 6\}$
 - C. $\{4, 5\}$
 - D. $\{1, 2, 3\}$
- 30.** The number of whole numbers which satisfy $x + 5 < 9$ is
- A. 3
 - B. 5
 - C. 4
 - D. 9
- 31.** If $2x + 3 \leq 11$ and x is a whole number, the greatest possible value of x is
- A. 4
 - B. 3
 - C. 5
 - D. 7
- 32.** The solution set of $x > -2$, where x is an integer between -4 and 3 , is
- A. $\{-3, -2, -1, 0\}$
 - B. $\{0, 1, 2\}$
 - C. $\{-2, -1, 0, 1, 2\}$
 - D. $\{-1, 0, 1, 2\}$
- 33.** On a number line, the inequality $x \geq 3$ is represented by
- A. an open dot at 3 and an arrow to the right
 - B. a solid dot at 3 and an arrow to the right
 - C. a solid dot at 3 and an arrow to the left
 - D. an open dot at 3 and an arrow to the left

- 34.** On a number line, an open dot at 2 with shading to the right represents
- A. $x > 2$
 - B. $x \geq 2$
 - C. $x < 2$
 - D. $x \leq 2$
- 35.** A student must score at least 40 marks to pass a test. If Kofi scores x marks and passes, then
- A. $x > 40$
 - B. $x < 40$
 - C. $x \geq 40$
 - D. $x \leq 40$
- 36.** A taxi can carry at most 4 passengers. If p passengers enter the taxi, the inequality representing the statement is
- A. $p > 4$
 - B. $p \geq 4$
 - C. $p < 4$
 - D. $p \leq 4$
- 37.** Adjoa has x oranges. She shares them equally among 5 friends and each friend receives at most 6 oranges. The inequality representing the statement is
- A. $x \leq 30$
 - B. $x \geq 30$
 - C. $x < 11$
 - D. $x \leq 6$
- 38.** The solution set of $x \leq 2$, where x is a whole number, is shown on a number line by shading the points
- A. 2 only
 - B. 0 and 1 only
 - C. 0, 1 and 2
 - D. all numbers less than 2
- 39.** A farmer has at most GH¢500.00 to spend on seedlings costing GH¢20.00 each. The greatest number of seedlings he can buy is
- A. 24
 - B. 25
 - C. 26
 - D. 50

- 40.** Amina's weight in kilograms, increased by 5, does not exceed 60. The greatest possible value of her weight w is
- A. 60
 - B. 65
 - C. 50
 - D. 55

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Translating word problems.

Question 2

(a) Solving linear inequalities.

Question 3

(a) Solution sets in given domains.

Question 4

(a) Number lines.

Question 5

(a) Real-life applications.

Question 6

(a) Integrated problems.